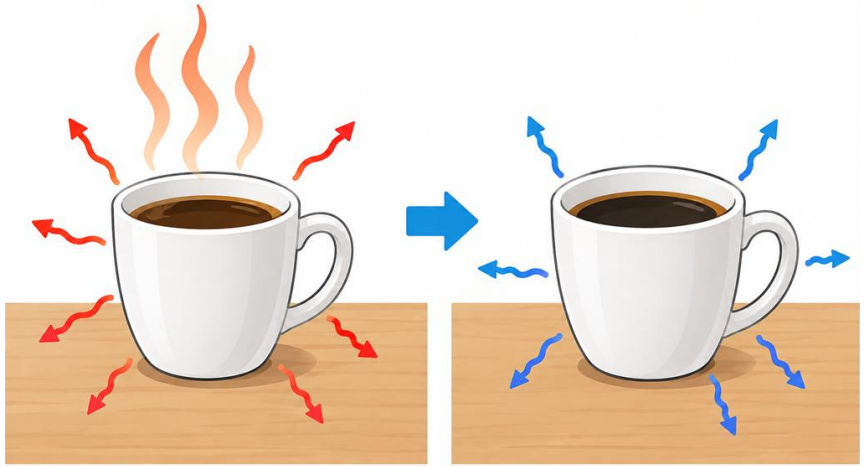
Chapter 5

Thermochemistry

- 5.1 The Nature of Chemical Energy
- 5.2 The First Law of Thermodynamics
- 5.3 Enthalpy
- 5.4 Enthalpies of Reaction
- 5.5 Calorimetry
- 5.6 Hess's Law
- 5.7 Enthalpies of Formation
- 5.8 Bond Enthalpies
- 5.9 Foods and Fuels

5.1 The Nature of Chemical Energy

Q. A cup of hot coffee left on your desk slowly cools down. Where does its energy go? Has any energy been destroyed?



Thermodynamics

Studies energy and its transformations!

Thermochemistry

Studies the relationship between chemical reactions and energy changes that involve heat.

5.1 The Nature of Chemical Energy

Energy, Work and Heat

Q. What is energy, and in what two ways can it be transferred?

- **Energy**: the capacity to do work or to transfer heat.
  - ✓ **Work (w)**: energy used to move an object against a force (**$w = F \times d$**).
  - ✓ **Heat (q)**: energy transferred from a hotter object to a colder one.

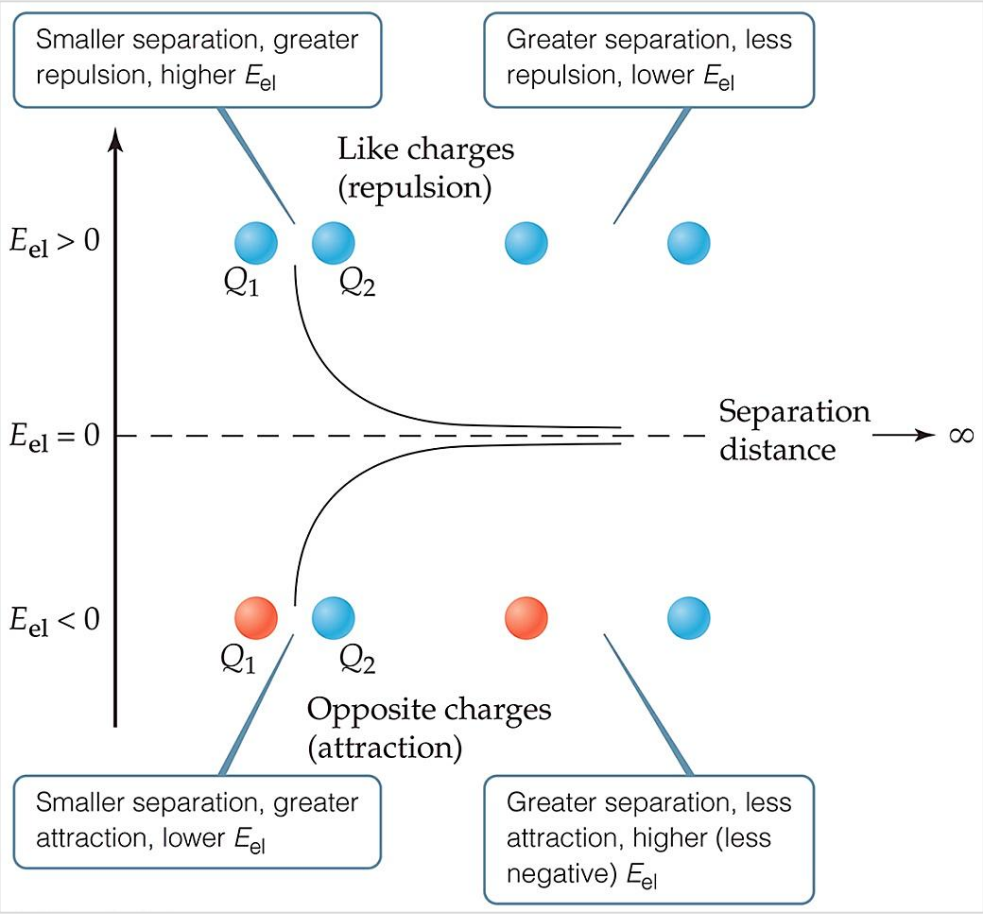
Potential energy, E_p	Kinetic energy, E_k
Energy of position or composition	Energy of motion
$E_p = mgh$	$E_k = \frac{1}{2}mv^2$
e.g. *a raised ball,* ***chemical bonds***	***e.g.*** *moving molecules, a rolling ball*

Q. Give one example of kinetic energy and one of potential energy in a chemical system.

5.1 The Nature of Chemical Energy

Chemical Energy Is Mainly Potential Energy

Q. How does the energy of two charged particles depend on their charges and their separation?



Electrostatic potential energy.

- The most important potential energy in chemistry is electrostatic potential energy (E_{el}).

$$E_{el} = \frac{kQ_1Q_2}{d}$$

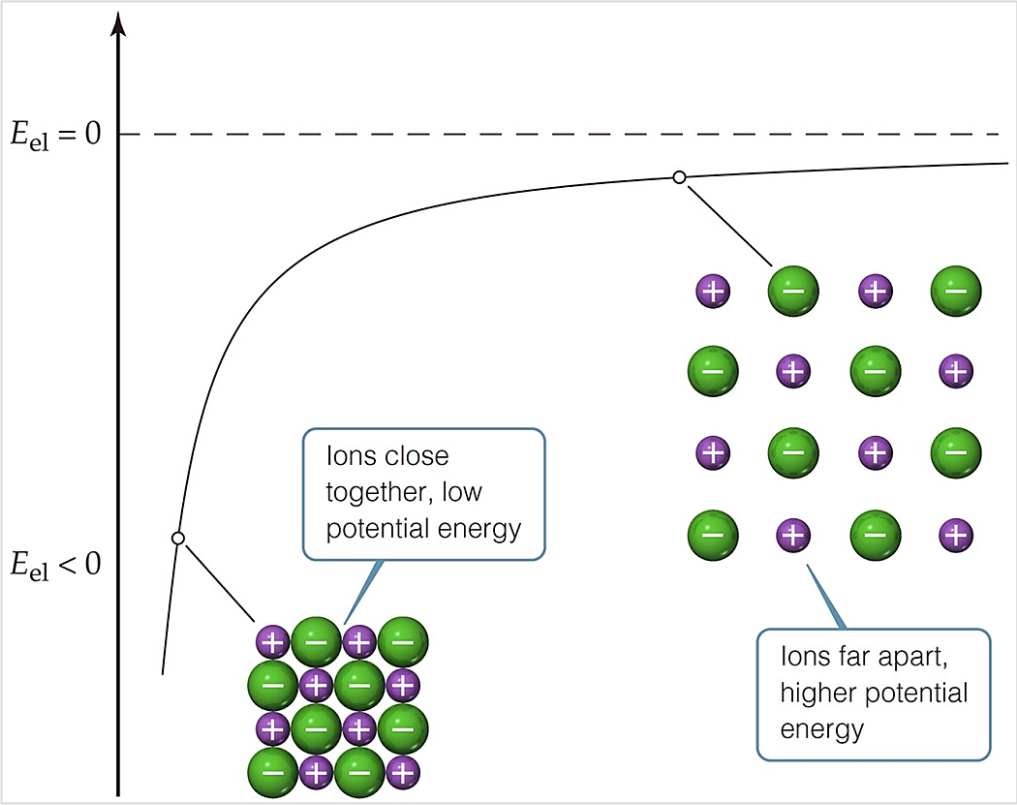
  - ✓ $\kappa = 8.99 \times 10^9$ J·m/C^2
  - ✓ Q_1, Q_2 = charges
  - ✓ d = distance

- Like charges repel: $E_{el} > 0$
- Opposite charges attract: $E_{el} < 0$
- As $d \rightarrow \infty$, $E_{el} \rightarrow 0$
- The lower (more negative) E_{el}, the more **stable** the system

5.1 The Nature of Chemical Energy

Attraction between Ions

Q. Is energy released or absorbed when oppositely charged ions come together? When they are pulled apart?



Electrostatic potential energy and ionic bonding

- Oppositely charged ions attract: bringing them closer lowers E_{el}.
- Ions close together → low potential energy → more stable.
  - ✓ Energy is released when chemical bonds are formed.
  - ✓ Energy is absorbed when chemical bonds are broken.

Q. Is energy absorbed or released when the H-H bond in H_2 is broken?

5.1 The Nature of Chemical Energy

Units of Energy

Q. Which energy units will we use, and how are they related?

Unit	Definition / conversion
▪ Joule (J) – SI unit	1 J = 1 kg·m^2/s^2
▪ Kilojoule (kJ)	1 kJ = 1000 J (used for reaction energies)
▪ Litre-atmosphere (L·atm) or 101.325 L.kPa	101.325 J (used for pressure-volume work)
▪ Calorie (cal)	1 cal = 4.184 J
▪ Nutritional Calorie (Cal)	1 Cal = 1000 cal = 1 kcal = 4.184 kJ

Example: A 2.0 kg object moving at 1.0 m/s has $E_k = \frac{1}{2}(2.0\ kg)(1.0\ m/s)^2 = 1.0\ kg{\cdot}m^2/s^2 = 1.0\ J$.

Q. A snack-bar label says 250 Cal. How many kilojoules is that?

Answer: 1.05×10^3 kJ

5.1 The Nature of Chemical Energy

Calculating Kinetic and Potential Energy

Q. A 5.4 kg bowling ball is lifted to a height of 16 m above the floor and dropped. (a) What is its potential energy (in J) at the top? (b) What is its kinetic energy just before it hits the floor? (c) What is its speed at that moment? ($g = 9.8\ m/s^2$)

a. $E_p = mgh =$ _____ $kg \cdot m^2/s^2$. (**Ans.** = 8.5×10^2 J)

b. Energy is conserved: all the potential energy becomes kinetic energy, so $E_k = 8.5 \times 10^2$ J just before impact.

c. $E_k = \frac{1}{2}mv^2 \rightarrow v =$ **?** (*Ans.* = 18 m/s)

5.1 The Nature of Chemical Energy

Q. Which has more kinetic energy: a 1.0 kg ball moving at 2.0 m/s or a 0.50 kg ball moving at 3.0 m/s?

Answer: $\frac{1}{2}(1.0)(2.0)^2$ = _____ J vs $\frac{1}{2}(0.50)(3.0)^2$ = ____ J.

Q. What is the electrostatic potential energy (in joules) between an electron and a proton that are separated by 230 pm? (*Ans.* -1.0×10^{-18} J). Does the potential energy of the two particles increase or decrease when the distance is increased to 1.0 nm?

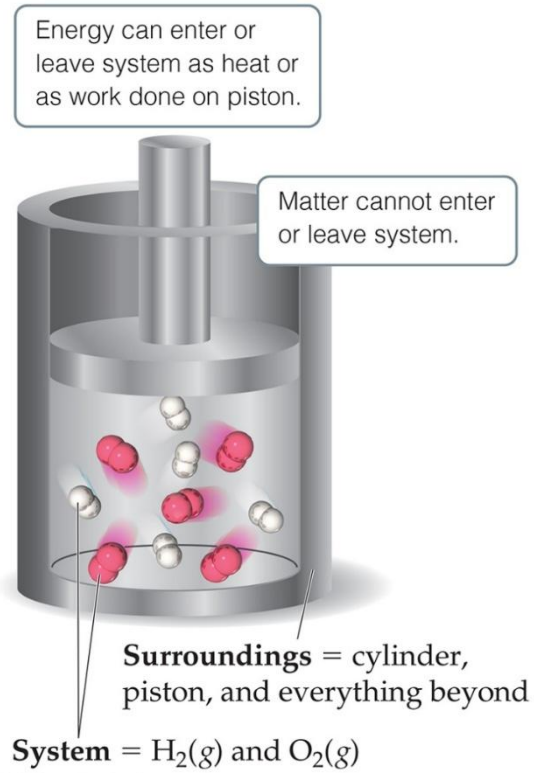
$$E_{el} = \frac{kQ_1Q_2}{d}$$

Q. A sodium ion, Na^+, with a charge of 1.6×10^{-19} C and a chloride ion, Cl^-, with charge of -1.6×10^{-19} C, are separated by a distance of 0.50 nm. How much work would be required to increase the separation of the two ions to an infinite distance? (*Ans.* 4.6×10^{-19} J).

5.2 The First Law of Thermodynamics

System and Surroundings

Q. When we study the reaction of H_2 and O_2 in a cylinder, what exactly are we studying?



A closed system.

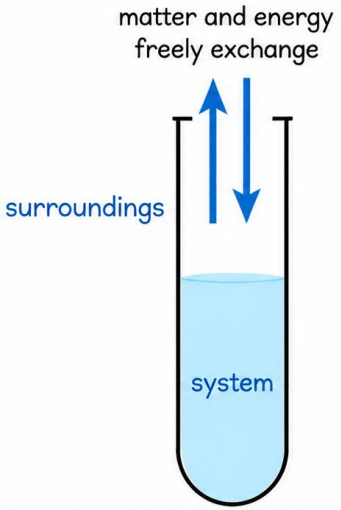
- **System**: the part of the universe we single out to study (here, the H_2 and O_2 molecules).
- **Surroundings**: everything else (the cylinder, the piston and everything beyond).
- Energy can cross the boundary as **heat** or as **work** (moving the piston).
- Matter cannot enter or leave here → a **closed system**.

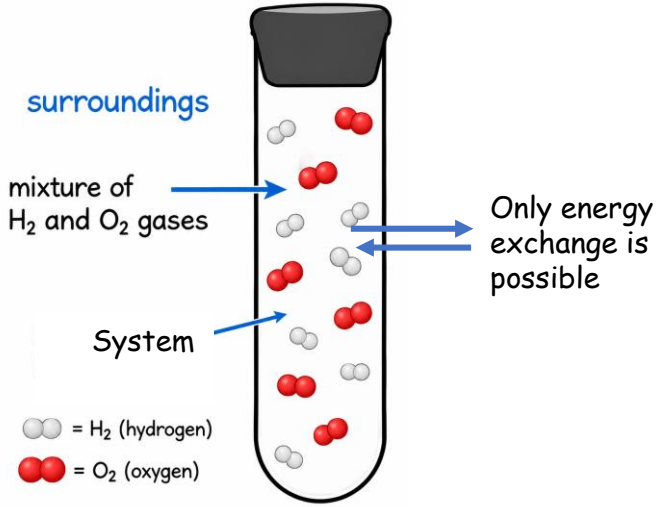
5.2 The First Law of Thermodynamics

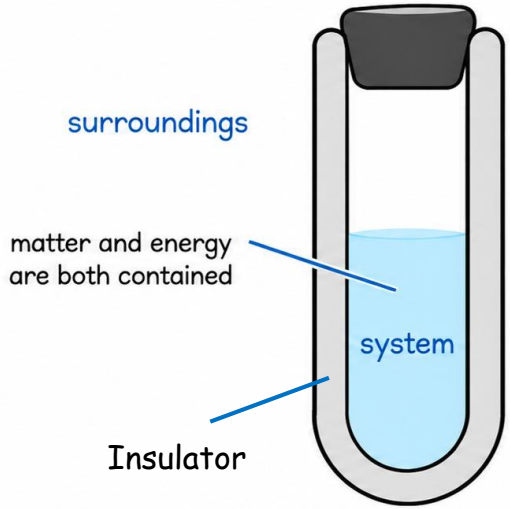
System and Surroundings: Types of Systems

Q. Can a system exchange both energy and matter with its surroundings?

Open system	Closed system	Isolated system
Exchanges energy and matter. Example: boiling water in an open test tube	Exchanges energy only Example: H_2 and O_2 in a cylinder /test tube with a piston	Exchanges neither matter nor energy. Example: Insulated thermos/test tube







5.2 The First Law of Thermodynamics

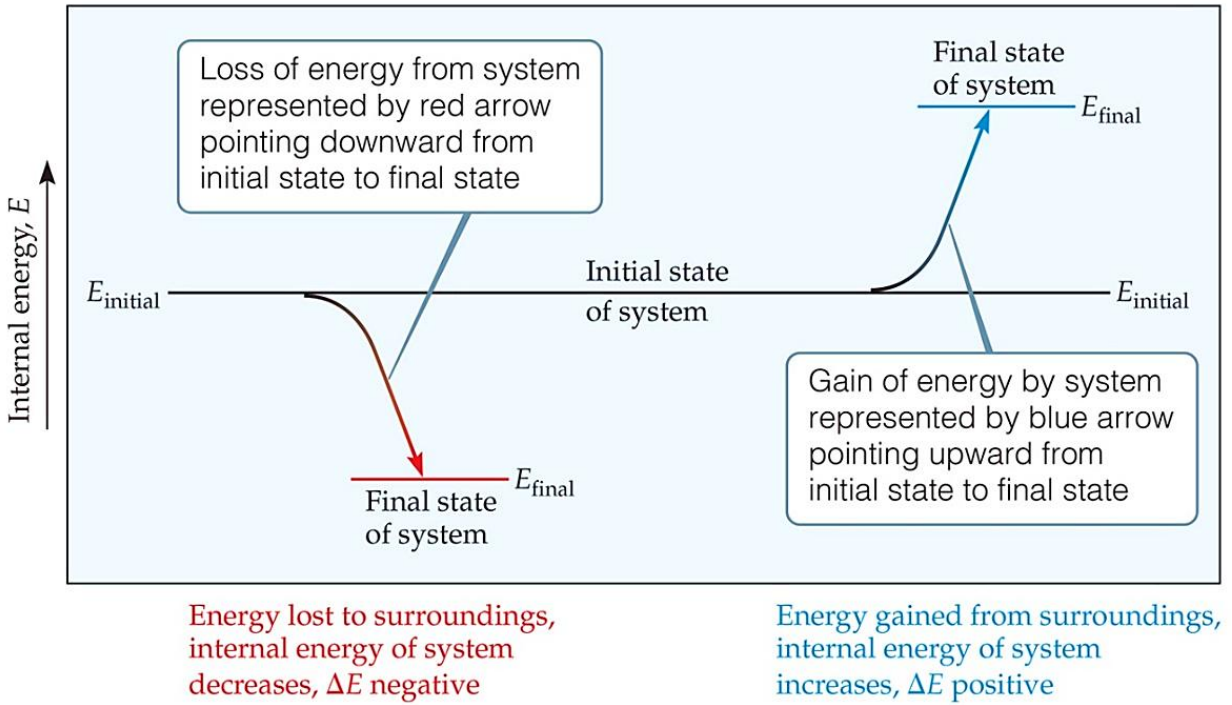
Internal Energy of a System (ΔE)

Q. What does the first law of thermodynamics tell us about energy?

First law of Thermodynamics: Energy can be converted from one form to another, but it is neither created nor destroyed.

Energy lost by system = Energy gained by surroundings

- **Internal energy, E**: the sum of all kinetic and potential energies of all components of the system.
- We cannot measure E itself, only its **change**: $\Delta E = E_{final} - E_{initial}$



Changes in internal energy.

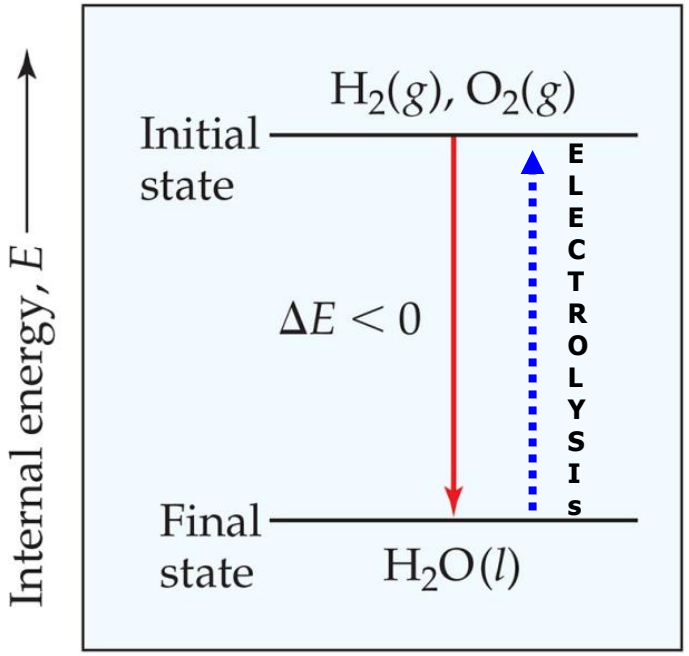
5.2 The First Law of Thermodynamics

Internal Energy: The Sign of ΔE

Every thermodynamic quantity has three parts: **a number, a unit, and a sign.**

- ✓ **$\Delta E = E_{final} - E_{initial}$**
- ✓ $\Delta E > 0$: $E_{final} > E_{initial}$ – the system gained energy from the surroundings.
- ✓ $\Delta E < 0$: $E_{final} < E_{initial}$ – the system lost energy to the surroundings.

Q. For $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$, is ΔE positive or negative? Is energy released or absorbed?



$E_{initial}$ greater than E_{final}; therefore, energy is released from system to surroundings during reaction and $\Delta E < 0$.

Energy diagram for the reaction $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$.

5.2 The First Law of Thermodynamics

Relating ΔE to Heat and Work

Q. In what **two forms can energy** be exchanged between a system and its surroundings?

$$\Delta E = q + w$$

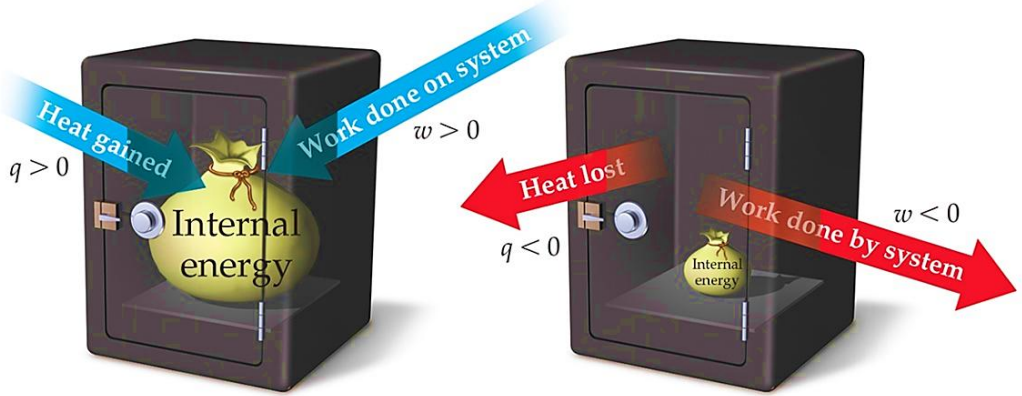
- ✓ q = heat absorbed or released by the system
- ✓ w = work done on or by the system
- Energy entering the system → positive
- Energy leaving the system → negative



System is interior of vault

Q. Can ΔE be negative even if the system absorbs heat?

Answer: Yes – if the system does more work on the surroundings than the heat it absorbs.



Energy deposited into system
ΔE > 0

Energy withdrawn from system
ΔE < 0

Sign conventions for heat and work.

5.2 The First Law of Thermodynamics

Relating ΔE to Heat and Work: Sign Conventions

Q. How do we decide the signs of q, w and ΔE?

TABLE 5.1 Sign Conventions for *q*, *w*, and *ΔE*

For *q*	+ means system *gains* heat	− means system *loses* heat
For *w*	+ means work done *on* system	− means work done *by* system
For *ΔE*	+ means *net gain* of energy by system	− means *net loss* of energy by system

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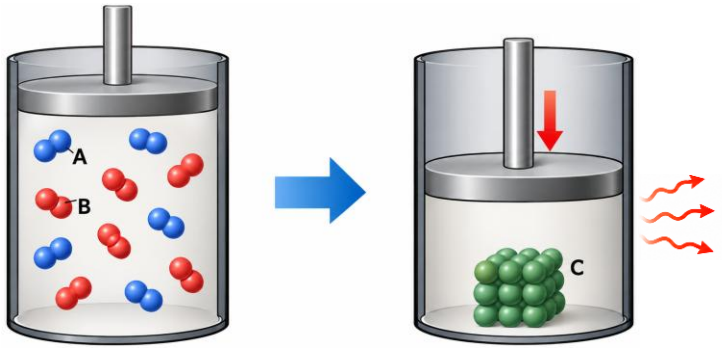
Always look at the process from the **system's** point of view: energy **in** = +, energy **out** = −.

Process	q	w	ΔE
Heat absorbed, work done on system	+	+	+
Heat released, work done by system	-	-	-
Heat absorbed, work done by system	+	-	depends on sizes
Heat released, work done on system	-	+	depends on sizes

5.2 The First Law of Thermodynamics

Relating ΔE to Heat and Work

Q. Gases A(g) and B(g) react in a cylinder fitted with a piston to form a solid, C(s). As the reaction occurs, the system loses 1150 J of heat to the surroundings. The piston moves down, and the surroundings do 480 J of work on the system. What is ΔE for the system?



Step 1. Heat is lost by the system → $q = -1150$ **J**

Step 2. Work is done on the system → $w = +480$ **J**

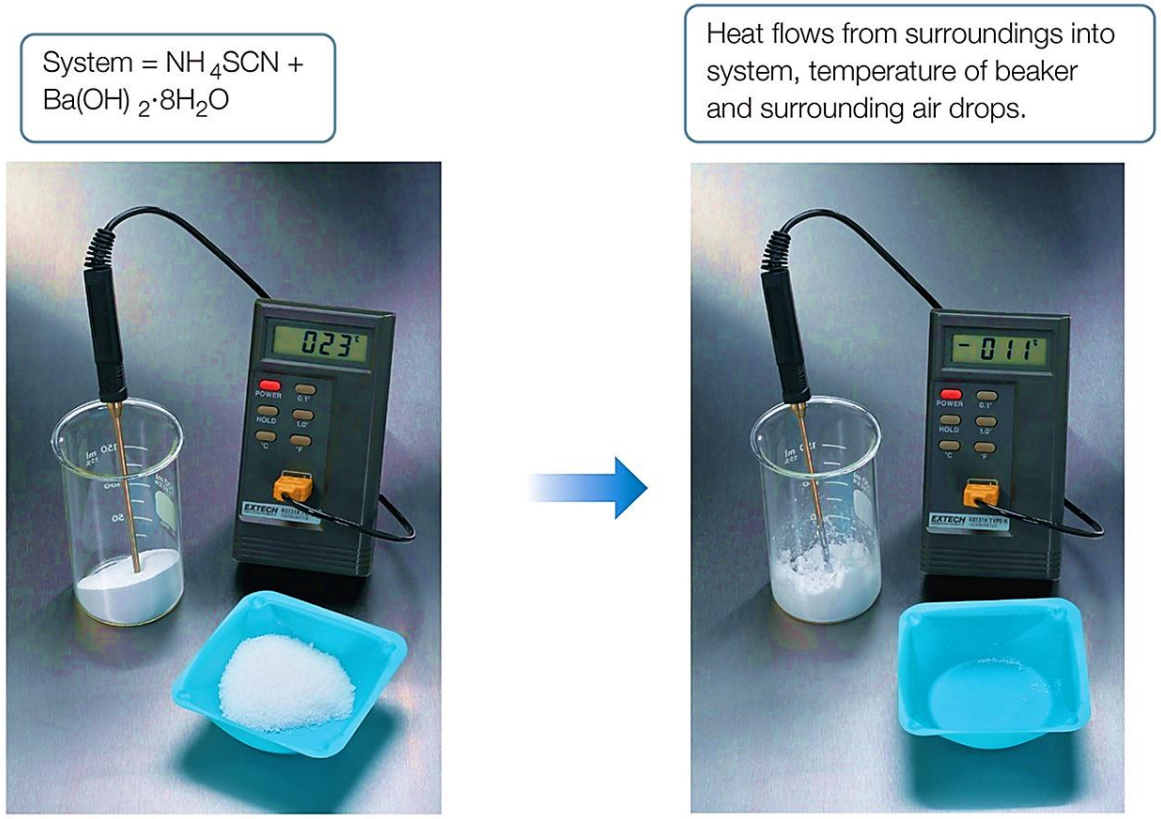
Step 3. $\Delta E = q + w = (-1150\ J) + (+480\ J) =$ **−670 J**

Step 4. $\Delta E < 0$: overall, the system has transferred 670 J of energy to the surroundings.

5.2 The First Law of Thermodynamics

Endothermic and Exothermic Processes

Q. $Ba(OH)_2 \cdot 8H_2O(s)$ is mixed with $NH_4SCN(s)$ and the thermometer reading falls sharply. Is heat flowing into or out of the system?



(a) $Ba(OH)_2 \cdot 8H_2O + 2\ NH_4SCN \longrightarrow Ba(SCN)_2 + 2\ NH_3 + 10\ H_2O$
An endothermic reaction

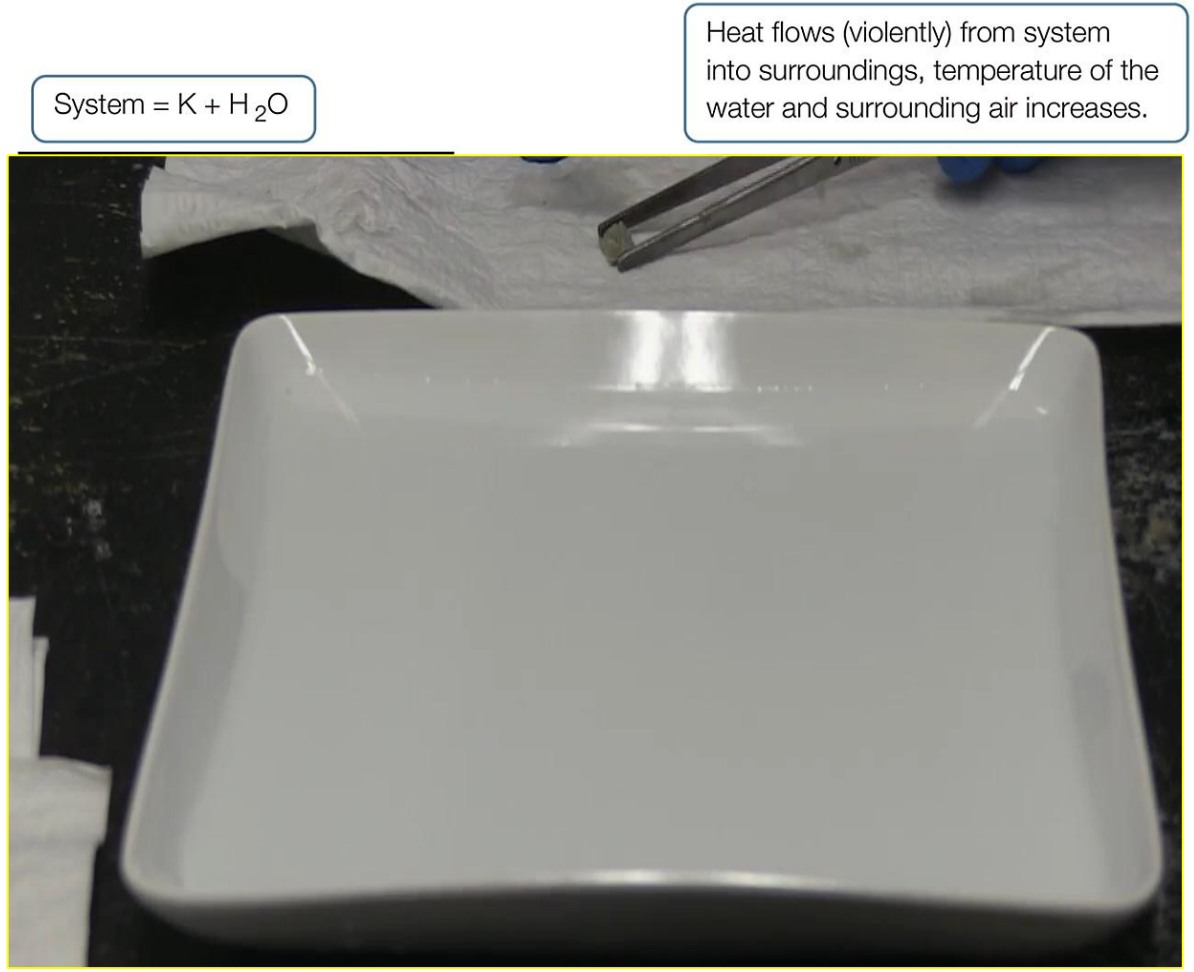
- **Endothermic**: heat is absorbed by the system from the surroundings (**q > 0**).
- The beaker and the air around it become colder.

Q. Identify the endothermic phase changes of water.

5.2 The First Law of Thermodynamics

Endothermic and Exothermic Processes

Q. Potassium reacts violently with water. Is heat flowing into or out of the system?



- **Exothermic**: heat is **released** by the system to the surroundings (**q < 0**).
- The water and the air around it become **hotter**.
- Combustion and neutralization are also exothermic

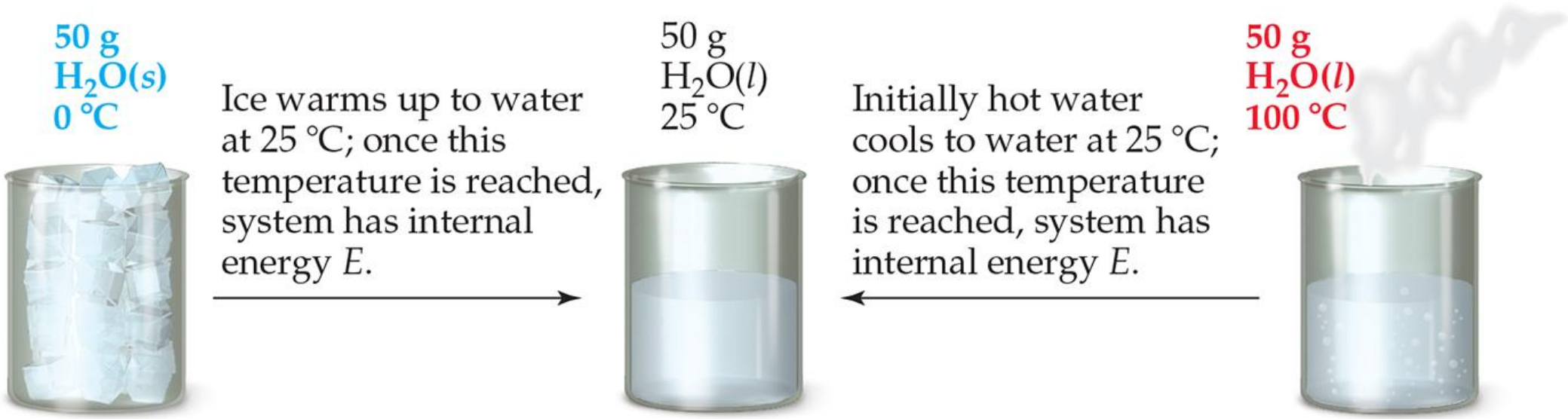
Q. Identify the exothermic phase changes of water.

5.2 The First Law of Thermodynamics

State Functions

Q. Does the internal energy of 50 g of water at 25 °C depend on how the water reached 25 °C?

- A **state function** depends only on the present state of the system, not on the path taken to reach it.
- Internal energy is a state function: ΔE depends only on $E_{initial}$ and E_{final}.
- Other state functions: P, V, T, and H (enthalpy).

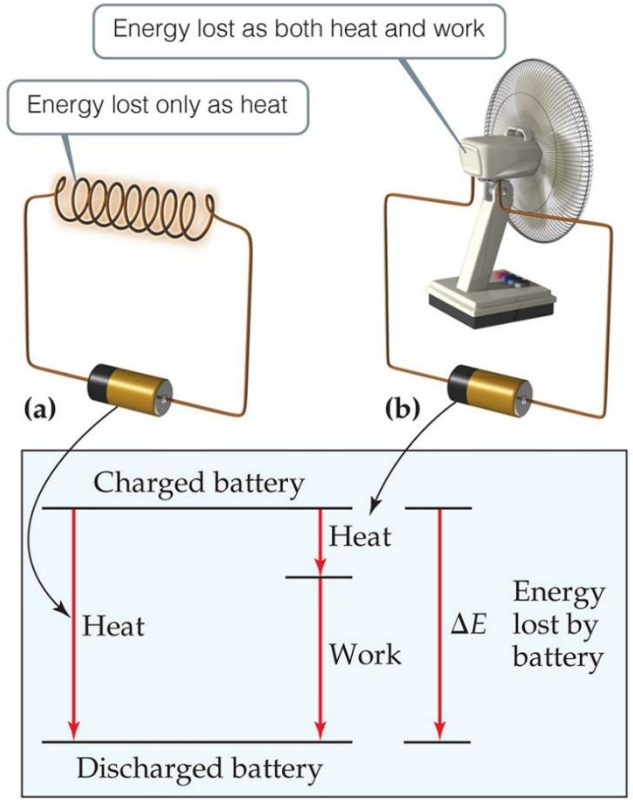


Internal energy, E, a state function

5.2 The First Law of Thermodynamics

State Functions: q and w Are Not State Functions

Q. A battery is discharged (a) through a wire or (b) by running a fan. Is ΔE the same? Are q and w the same?



Internal energy is a state function, but heat and work are not.

- ΔE of the battery is the **same** in both cases (same initial and final states).

✓ (a) Short circuit: the energy is lost only as **heat**; w = 0.

✓ (b) Fan: the energy is lost as **work** plus some heat.

- q and w depend on the path: they are **not** state functions.

5.2 The First Law of Thermodynamics

Q. A system absorbs 50 kJ of heat at constant volume (w = 0). What is ΔE? Is the process endothermic or exothermic?

Answer: +50 kJ; heat is absorbed → ?.

Q. Calculate ΔE and determine whether the process is endothermic or exothermic for the following cases:

(a) q = 0.763 kJ and w =-840 J. (*Ans.* ΔE = -0.077 kJ, Endothermic)

(b) A system releases 66.1 kJ of heat to its surroundings while the surroundings do 44.0 kJ of work on the system (*Ans.* ΔE = –22.1 kJ, exothermic).

5.3 Enthalpy (H)

Pressure–Volume Work

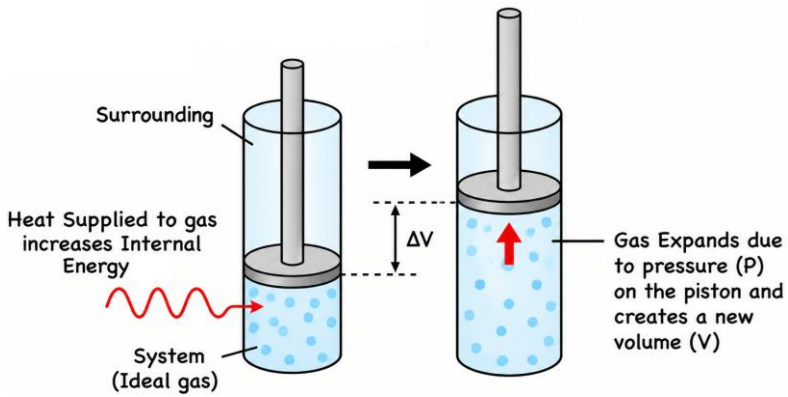
Q. Consider a reaction carried out in an open beaker at constant atmospheric pressure. If the reaction produces a gas, what extra job does that gas do?

Answer: It pushes back the atmosphere – it does **pressure–volume work** on the surroundings. Enthalpy takes this into account.

- The work involved in the expansion or compression of gases is called pressure-volume work (P–V work)
- Enthalpy (**H**) is the internal energy plus the product of the pressure, P, and volume, V, of the system.

$$H = E + PV$$

- Because energy, pressure, and volume are all state functions, enthalpy is also a **state function.**

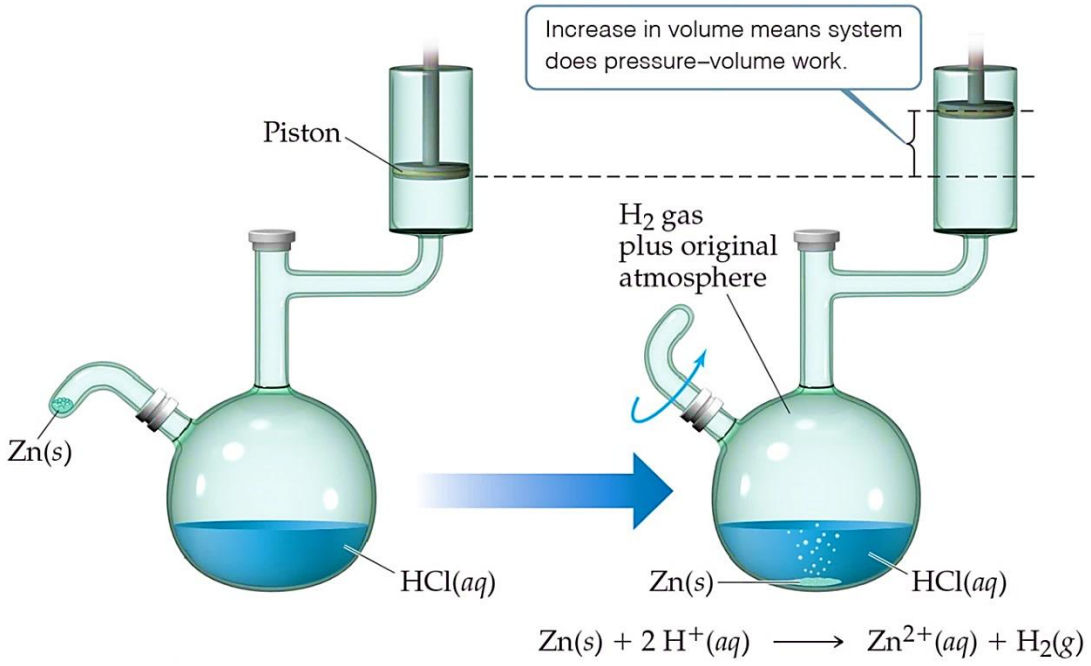


5.3 Enthalpy (H)

Pressure–Volume Work: Work Done by a Reaction

Q. Zn reacts with HCl(aq) in a flask fitted with a piston. What is the sign of **w** as H_2 gas forms?

$$Zn(s) + 2H^+(aq) \rightarrow Zn^{2+}(aq) + H_2(g)$$



A system that does work on its surroundings

- ✓ H_2 gas pushes the piston up against atmospheric pressure → $\Delta V > 0$.
- ✓ **$w = -P\Delta V$** (When a gas expands, the system does work on the surroundings).
- ✓ $\Delta V = V_{final} - V_{initial}$

$w = -P\Delta V$
For any process occurring at constant external pressure

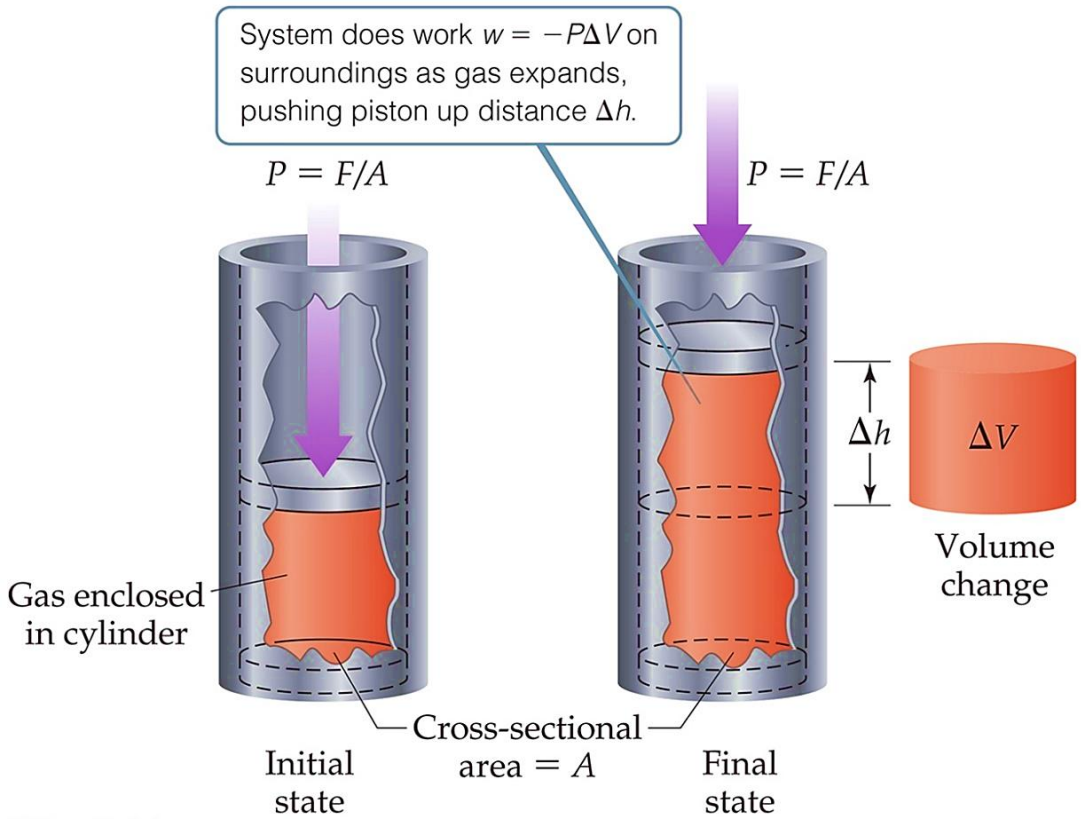
What is the unit of w?

$PV = (F/A)V = (kg\ \ m/s^2.m^2)\ m^3 = (kg.m^2)/(s^2) = J$

5.3 Enthalpy (H)

Pressure–Volume Work

Q. What kind of work is usually done in chemical and physical changes?



- Usually the only work is **mechanical work** from a change in gas volume.
- For a gas expanding against a constant external pressure P:

$$w = -P\Delta V$$

- ✓ Expansion ($\Delta V > 0$) → $w < 0$: work is done **by** the system.
- ✓ Compression ($\Delta V < 0$) → $w > 0$: work is done **on** the system.
- ✓ Units: 1 kPa·L = 1 J (1 kPa·L = 1 J; 1 L·atm = 101.3 J)

5.3 Enthalpy (H)

Pressure–Volume Work

Q. A fuel is burned in a cylinder equipped with a piston. The initial volume of the cylinder is 0.250 L, and the final volume is 0.980 L. If the piston expands against a constant pressure of 136.8 kPa, how much work (in J) is done?

Step 1. $\Delta V = V_{final} - V_{initial} = 0.980\ L - 0.250\ L = 0.730\ L$

Step 2. $w = -P\Delta V = -(136.8\ kPa)(0.730\ L) = \mathbf{-99.9\ kPa \cdot L}$

Step 3. $1\ kPa \cdot L = (10^3\ Pa)(10^{-3}\ m^3) = \mathbf{1\ Pa \cdot m^3 = 1\ J} \rightarrow \mathbf{w = -99.9\ J}$

Step 4. $w < 0$: the expanding gas does work **on** the surroundings.

5.3 Enthalpy

Enthalpy Change (ΔH)

Q. Why do chemists use enthalpy rather than internal energy for most reactions?

- Most reactions occur at constant pressure, where P–V work can occur.
- Enthalpy is defined as:

$$H = E + PV$$

- H is a state function (E, P and V are all state functions).
- At constant pressure: $\Delta H = \Delta E + P\Delta V$

Derivation (constant P):

$\mathbf{\Delta H = \Delta E + P\Delta V}$

$\Delta E = q_P + w$ and $P\Delta V = -w$

$\Delta H = (q_P + w) - w$

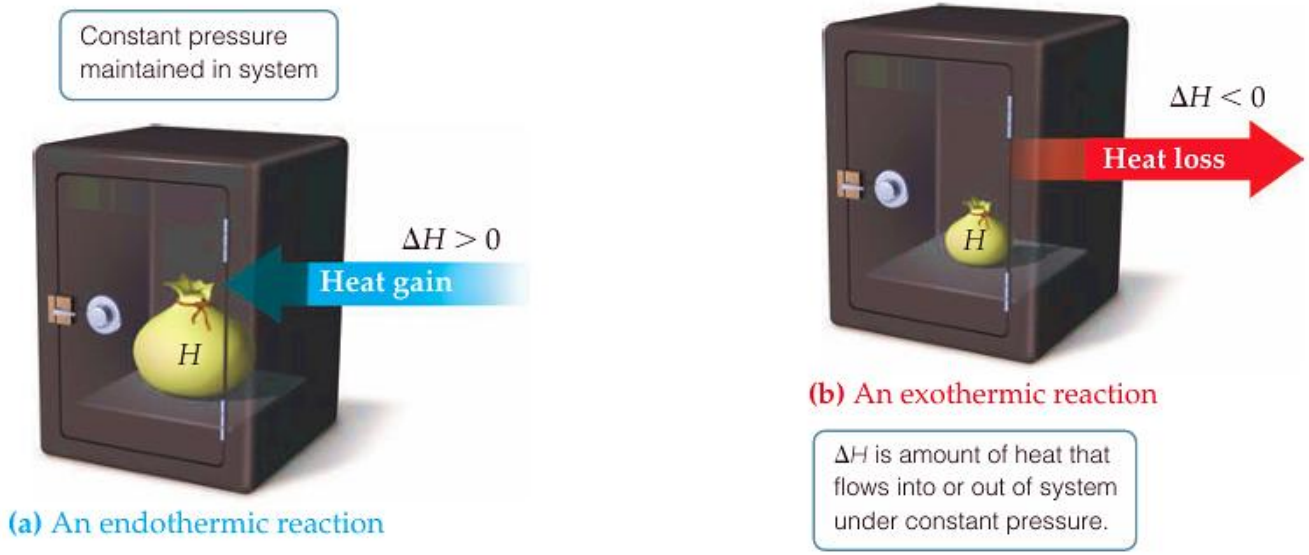
$\mathbf{\Delta H = q_P}$

At constant pressure, the change in enthalpy equals the heat gained or lost at constant pressure ($\Delta H = \mathbf{q_P}$).

5.3 Enthalpy

Enthalpy Change (ΔH): The Sign of ΔH

Q. What does the sign of ΔH tell us?



- **ΔH > 0**: heat flows **into** the system → **endothermic**.
- **ΔH < 0**: heat flows **out of** the system → **exothermic**.
- ΔH is easy to measure (it is simply the heat flow at constant pressure).

5.3 Enthalpy

Enthalpy Change (ΔH): The Sign of ΔH

Q. Indicate the sign of the enthalpy change, ΔH, in the following processes carried out under atmospheric pressure, and indicate whether each process is endothermic or exothermic: (a) an ice cube melts; (b) 1 g of butane (C_4H_{10}) is combusted in sufficient oxygen to give complete combustion to CO_2 and H_2O.

(a) Melting requires heat from the surroundings → **ΔH > 0, endothermic.**

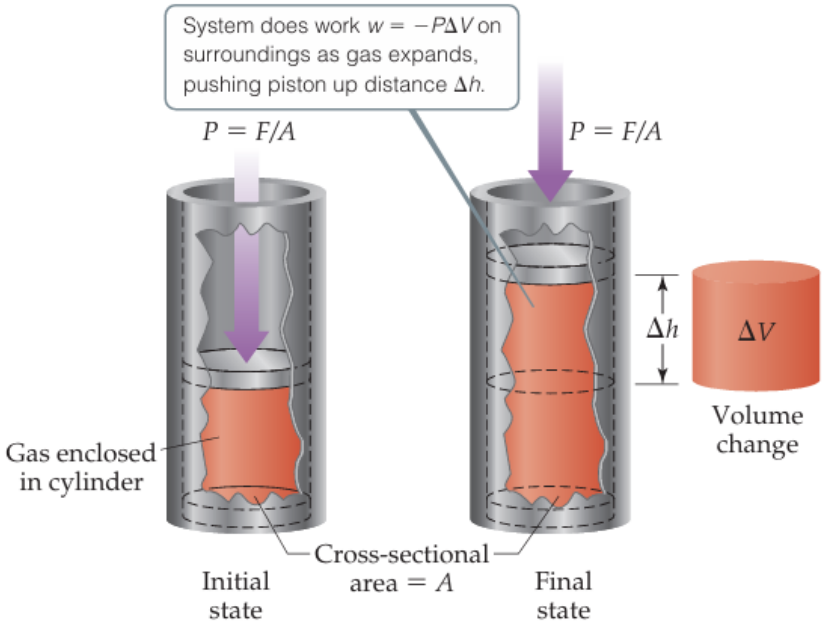
(b) Combustion releases heat to the surroundings → **ΔH < 0, exothermic.**

- ✓ At constant pressure $\Delta H = q_P$
- ✓ Heat absorbed by the system → ΔH positive
- ✓ Heat released → ΔH negative

5.3 Enthalpy

Summary point: Energy, Enthalpy, and P–V Work

Q. Where does w = -PΔV come from, and how are the heats measured at constant volume and at constant pressure related to ΔE and ΔH?



- A gas is confined in a cylinder by a piston of area A; the pressure on the gas is P = F/A, so F = P × A.
- When the gas expands and pushes the piston up a distance h: |w| = F × h = P × A × h = PΔV (ΔV = A × h).
- The system does work on the surroundings, so **w = –PΔV** and **ΔE = q + w = q – PΔV**.
- For most reactions PΔV is small, so ΔH ≈ ΔE; the difference matters only when the number of moles of gas changes.

Constant pressure

$q_P = \Delta E + P\Delta V \rightarrow \mathbf{\Delta H = q_P}$

e.g. an open beaker, a coffee-cup calorimeter

Constant volume (ΔV = 0)

$w = 0 \rightarrow \mathbf{\Delta E = q_V}$

e.g. a bomb calorimeter

5.3 Enthalpy

Summary point: Energy, Enthalpy, and P–V Work

Q. When are ΔH and ΔE very different?

- ΔH = ΔE + PΔV: the two differ only by the P–V work term.
- Reactions of only solids and liquids: ΔV ≈ 0, so **ΔH ≈ ΔE**.
- Reactions in which the number of moles of gas changes: PΔV can be significant.
- Even then, PΔV is usually small compared with ΔH, so ΔH ≈ ΔE for most reactions.

Q. For which reaction will ΔH and ΔE differ the most?

(a) $NaCl(s) \rightarrow Na^+(aq) + Cl^-(aq)$

(b) $C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l)$

(c) $H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$

Answer: (b): 6 mol of gas → 3 mol of gas, a large ΔV. (a) involves no gas; in (c) the moles of gas do not change.

Q. A reaction carried out at constant pressure releases 100 kJ of heat. What is ΔH?

Answer: $\Delta H = q_P = -100$ kJ (exothermic).

5.4 Enthalpies of Reaction

Enthalpy of Reaction

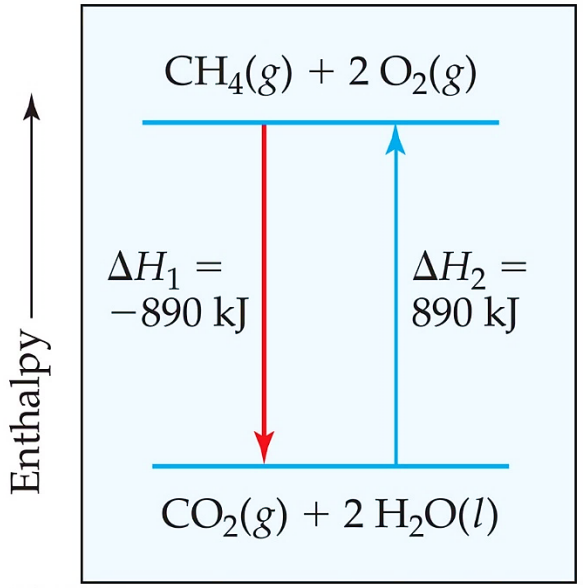
Q. Burning 1 mol of CH_4 releases 890 kJ of heat. How much heat is released by burning 2 mol? What is ΔH if the reaction is reversed?

Answer: 2 mol releases 1780 kJ (ΔH = -1780 kJ). For the reverse reaction, ΔH = +890 kJ (heat is absorbed).

- **$\Delta H_{rxn} = H_{products} - H_{reactants}$**
- ΔH_{rxn} is called the enthalpy (heat) of reaction.
- A thermochemical equation gives the balanced equation together with its ΔH:

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l) \quad \Delta H = -890 \text{ kJ}$$

- In an enthalpy diagram, the products lie below the reactants for an exothermic reaction (and above for an endothermic one).



ΔH for a reverse reaction.

5.4 Enthalpies of Reaction

Enthalpy of Reaction: Thermochemical equations

- Represents a balanced chemical equation.
- The physical states should be shown.
- The energy associated with the reaction should be written to the right.

$H_2(g) + 1/2\ O_2(g) \rightarrow H_2O(g) \qquad \Delta H = -242\ kJ$

$H_2(g) + 1/2\ O_2(g) \rightarrow H_2O(l) \qquad \Delta H = -286\ kJ$

$S(s) + O_2(g) \rightarrow SO_2(g) \qquad \Delta H = -296.8\ kJ$

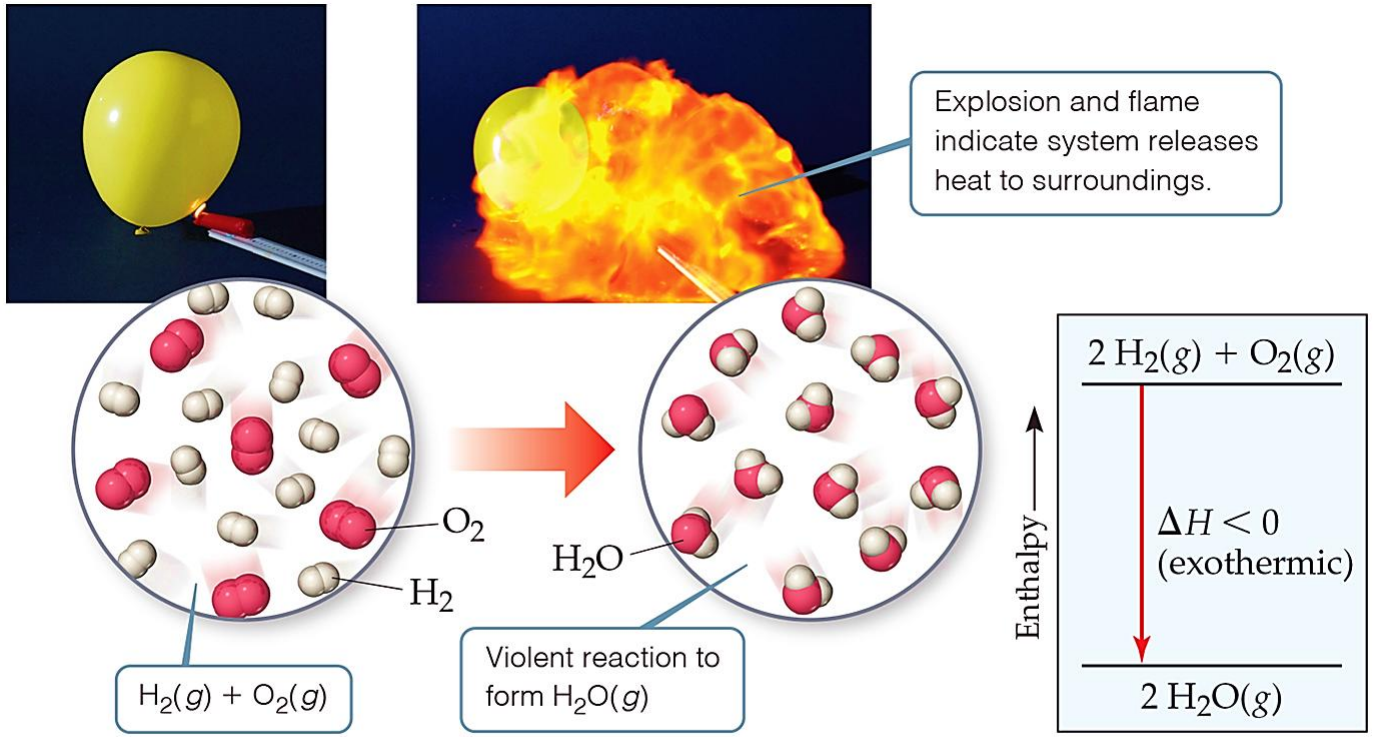
$Mg(s) + 12\ O_2(g) \rightarrow MgO(s) \qquad \Delta H = -601.6\ kJ$

$H_2O\ (s) \rightarrow H_2O\ (l) \qquad \Delta H = 6.00\ kJ$

5.4 Enthalpies of Reaction

Example: Combustion of Hydrogen

Q. Is the reaction of H_2 with O_2 endothermic or exothermic? How can you tell from the enthalpy diagram?



Exothermic reaction of hydrogen with oxygen

- $2H_2(g) + O_2(g) \rightarrow 2H_2O(g)$
  $\Delta H = -483.6$ kJ
- Products are lower in enthalpy → exothermic.
- ΔH refers to the amounts as written: 483.6 kJ is released per 2 mol H_2.

5.4 Enthalpies of Reaction

Guidelines for Thermochemical Equations

Q. How does ΔH change when we change the amount, the direction or the physical states?

1. ΔH is extensive

ΔH is proportional to the amount of reactant consumed.

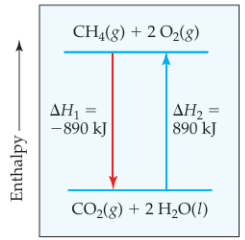
$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O(l)$
$\Delta H = -890$ kJ

$2CH_4 + 4O_2 \rightarrow 2CO_2 + 4H_2O(l)$
$\Delta H = -1780$ kJ

2. Reverse → change sign

$\Delta H_{reverse} = -\Delta H_{forward}$

$CO_2 + 2H_2O(l) \rightarrow CH_4 + 2O_2$
$\Delta H = +890$ kJ



3. States matter

ΔH depends on the states of reactants and products.

$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O(l)$
$\Delta H = -890$ kJ

$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O(g)$
$\Delta H = -802$ kJ

Why -890 vs -802 kJ? Condensing 2 mol of $H_2O(g)$ to $H_2O(l)$ releases an extra 88 kJ.

5.4 Enthalpies of Reaction

Using ΔH as a Conversion Factor

Q. How much heat is released when 4.50 g of methane gas is burned in a constant-pressure system?

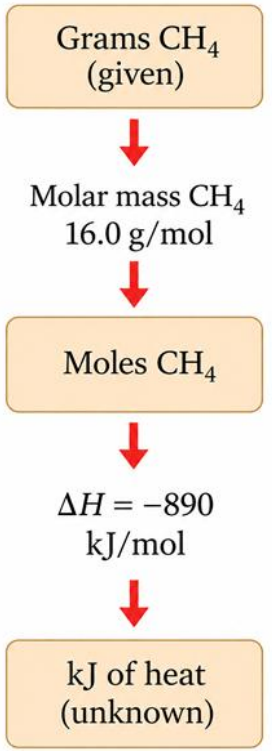
$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$ **$\Delta H = -890$ kJ**

Step 1. Moles of CH_4: 4.50 g × (1 mol / 16.04 g) = 0.281 mol CH_4

Step 2. From the equation: ΔH = -890 kJ per 1 mol CH_4

Step 3. q = 0.281 mol CH_4 × (-890 kJ / 1 mol CH_4) = -250 kJ

Step 4. The negative sign means 250 kJ of heat is released to the surroundings.



Q. $C(s) + O_2(g) \rightarrow CO_2(g)$, ΔH = -393.5 kJ. How much heat is released when 24.0 g of carbon burns?

Answer: Ans. 787 kJ released

5.4 Enthalpies of Reaction

Using Enthalpy as a Guide

Q. Does a negative ΔH guarantee that a reaction will happen by itself?

- A **spontaneous process** is thermodynamically favored (a brick falls when released); thermodynamics says nothing about its **rate.**
- Highly exothermic reactions usually proceed spontaneously, but may need energy to get started: $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(g)$, $\Delta H = -242$ kJ – H_2 and O_2 can coexist indefinitely until ignited.
- ΔH is not a foolproof guide: ice melting is endothermic ($\Delta H = +6.01$ kJ) yet spontaneous above 0 °C, while water freezes spontaneously below 0 °C.
- Rule of thumb: when ΔH is large (about 100 kJ or more) it dominates – large negative $\Delta H \rightarrow$ spontaneous; large positive $\Delta H \rightarrow$ spontaneous only in reverse. (More in Chapter 19.)

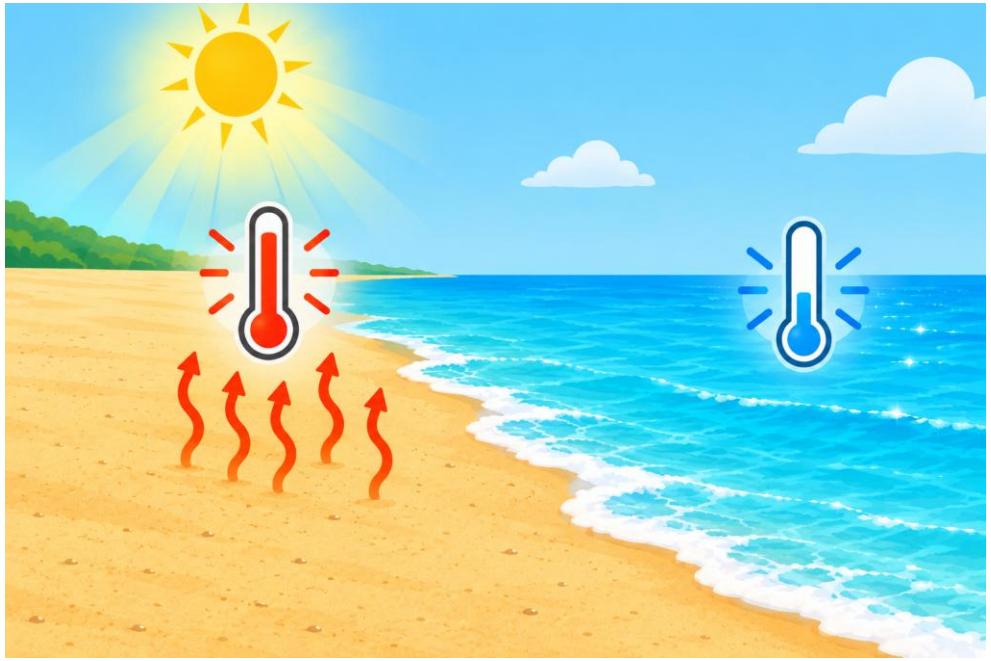
Q. Which is likely to be spontaneous: $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$, $\Delta H = -890$ kJ, or its reverse?

Answer: The forward combustion – ΔH is large and negative.

5.5 Calorimetry

Measuring Heat Flow

Q. On a sunny day at the beach, the sand becomes too hot to walk on, but the sea water stays cool. Why?

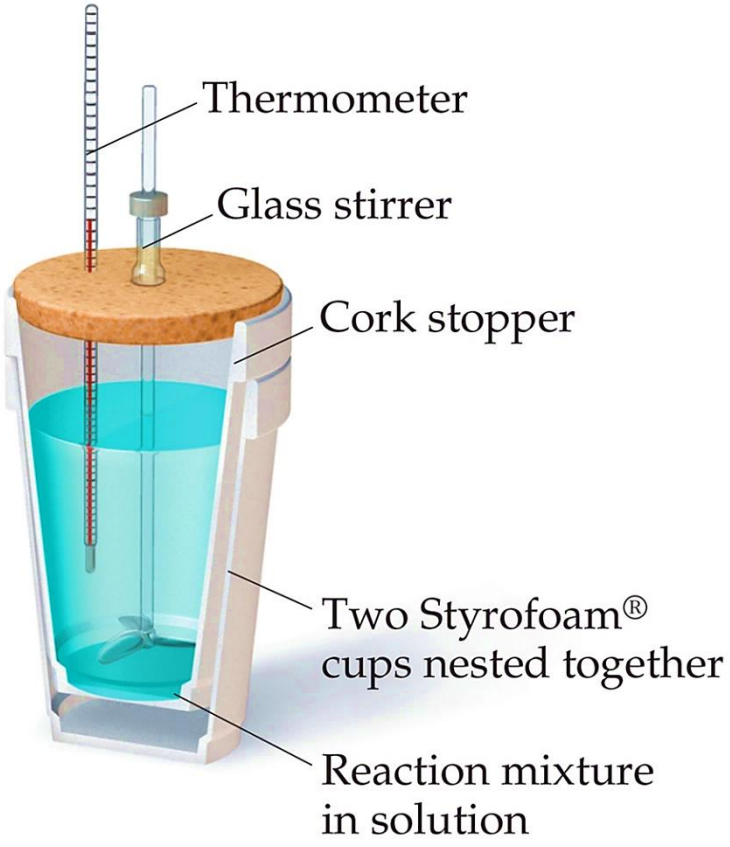


Answer: Water has a much higher specific heat than sand: the same amount of heat produces a much smaller temperature rise in water.

5.5 Calorimetry

Measuring Heat Flow

Q. We cannot measure absolute enthalpies. How can we measure ΔH?



Coffee-cup calorimeter.

- **Calorimetry**: the measurement of heat flow.
- **Calorimeter**: the device used to measure heat flow.
- Idea: the heat released or absorbed by a reaction changes the temperature of the surroundings (usually water) by a measurable amount.
- Two common types: **coffee-cup** (constant pressure) and **bomb** (constant volume) calorimeters.

5.5 Calorimetry

Heat Capacity and Specific Heat

Q. How much heat is needed to raise the temperature of a substance?

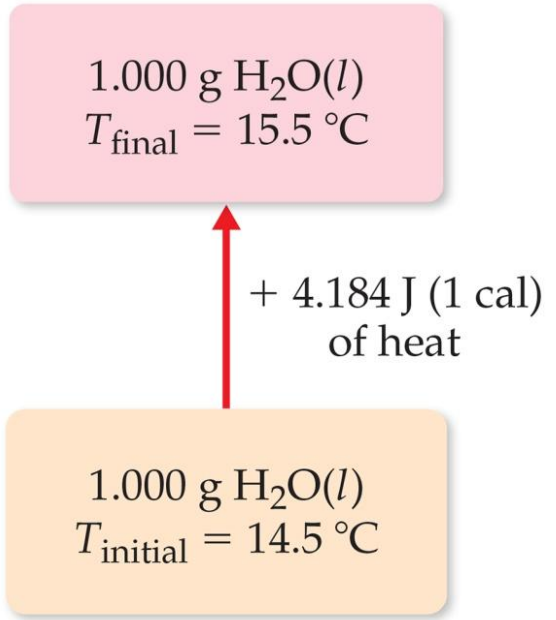
- Heat capacity (C): heat needed to raise the temperature of an object by 1 K (or 1 °C).
- Molar heat capacity (C_m): heat capacity of 1 mol of a substance (J/mol·K).
- Specific heat (C_s): heat capacity of 1 g of a substance (J/g·K).

$$\textit{Specific heat } (C_s) = \frac{(\textit{quantity of heat transferred})}{(\textit{grams of substance})\ x\ (\textit{temperature chnage})} = \frac{q}{m\ x\ \Delta T}$$

- Heat transferred: $\mathbf{q = C_s \times m \times \Delta T}$

TABLE 5.2 Specific Heats of Some Substances at 298 K

Elements		Compounds	
Substance	Specific Heat (J/g-K)	Substance	Specific Heat (J/g-K)
$N_2(g)$	1.04	$H_2O(l)$	4.18
$Al(s)$	0.90	$CH_4(g)$	2.20
$Fe(s)$	0.45	$CO_2(g)$	0.84
$Hg(l)$	0.14	$CaCO_3(s)$	0.82



Specific heat of water

5.5 Calorimetry

Heat Capacity and Specific Heat

Q. (a) How much heat is needed to warm 250 g of water (about 1 cup) from 22 °C to 98 °C? (b) What is the molar heat capacity of water? (C_s = 4.18 J/g·K)

Step 1. ΔT = 98 °C – 22 °C = 76 °C = 76 K (a change of 1 °C equals a change of 1 K)

Step 2. $q = C_s \times m \times \Delta T$ = (4.18 J/g·K)(250 g)(76 K) = **7.9×10^4 J**

Step 3. $C_m = C_s \times$ molar mass = (4.18 J/g·K)(18.0 g/mol) = **75.2 J/mol·K**

5.5 Calorimetry

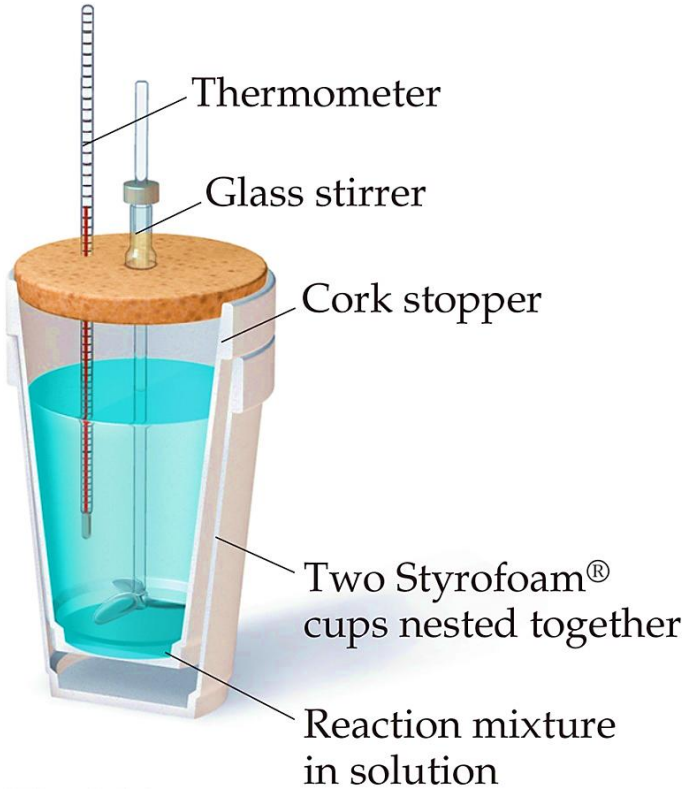
Constant-Pressure Calorimetry: Coffee-Cup Calorimeter

Q. How can a simple coffee-cup calorimeter give us ΔH for a reaction in solution?

- The reaction occurs in solution at constant (atmospheric) pressure, so the heat measured is ΔH.
- Heat released by the reaction is absorbed by the solution (no heat lost to the cup or air):

$$q_{rxn} = -q_{soln} = -(C_s \times m_{soln} \times \Delta T)$$

  - Dilute solutions: $C_s \approx 4.18$ J/g·K, density ≈ 1.0 g/mL.
    - T **rises** → exothermic (ΔH < 0)
    - T **falls** → endothermic (ΔH > 0)
  - ΔH (kJ/mol) = q_{rxn} ÷ moles reacted.



Coffee-cup calorimeter.

5.5 Calorimetry

Constant-Pressure Calorimetry

Q. A student mixes 50 mL of 1.0 M HCl and 50 mL of 1.0 M NaOH in a coffee-cup calorimeter. The temperature rises from 21.0 °C to 27.5 °C. Calculate ΔH in kJ/mol HCl. (Total volume 100 mL, density 1.0 g/mL, C_s = 4.18 J/g·K, no heat lost.)

Step 1. m_{soln} = 100 mL × 1.0 g/mL = 100 g; ΔT = 27.5 – 21.0 = 6.5 K

Step 2. q_{soln} = (4.18 J/g·K)(100 g)(6.5 K) = 2.7 × 10^3 J = 2.7 kJ

Step 3. $q_{rxn} = -q_{soln}$ = –2.7 kJ (exothermic)

Step 4. mol HCl = (0.050 L)(1.0 mol/L) = 0.050 mol → ΔH = –2.7 kJ ÷ 0.050 mol = **–54 kJ/mol**

5.5 Calorimetry

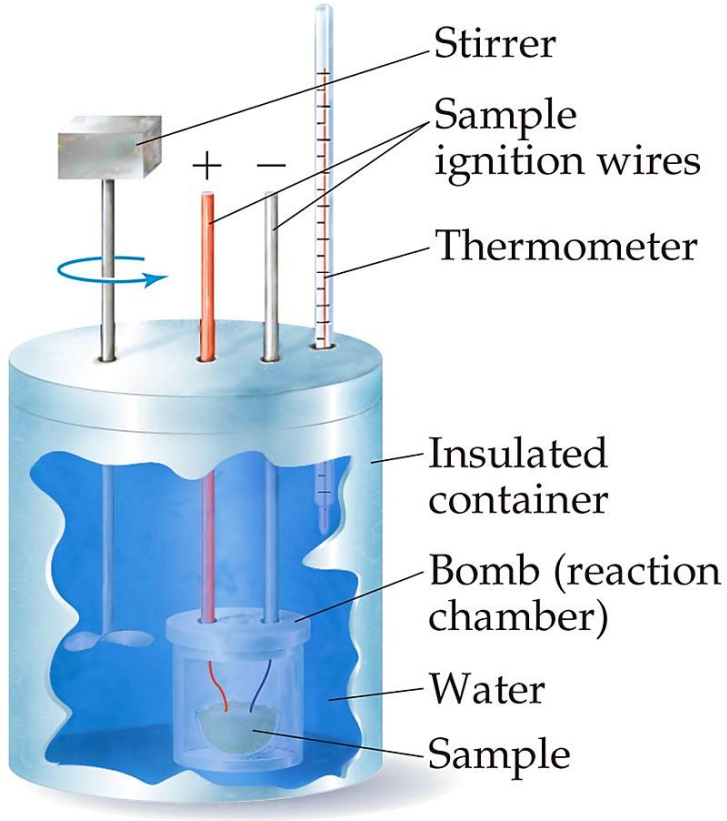
Bomb Calorimetry (Constant-Volume Calorimetry)

Q. How is the heat of combustion of a fuel or a food measured?

- The sample is burned in pure O_2 inside a sealed steel "bomb" immersed in water.
- The heat released raises the temperature of the whole calorimeter:

$$q_{rxn} = -C_{cal} \times \Delta T$$

- C_{cal} = heat capacity of the whole calorimeter (kJ/°C), found by calibration.
- The volume is constant → we measure **ΔE**, not ΔH. For most reactions the difference is very small.



Bomb calorimeter.

5.5 Calorimetry

Bomb Calorimetry (Constant-Volume Calorimetry)

Q. Methylhydrazine (CH_6N_2) is used as a rocket fuel. When 4.00 g of it is burned in a bomb calorimeter, the temperature rises from 25.00 °C to 39.50 °C. C_{cal} = 7.794 kJ/°C. Calculate the heat of reaction per mole of CH_6N_2.

Step 1. $\Delta T = 39.50\ °C - 25.00\ °C = 14.50\ °C$

Step 2. $q_{rxn} = -C_{cal} \times \Delta T = -(7.794\ kJ/°C)(14.50\ °C) = -113.0\ kJ$

Step 3. $mol\ CH_6N_2 = 4.00\ g \div 46.1\ g/mol = 0.0868\ mol$

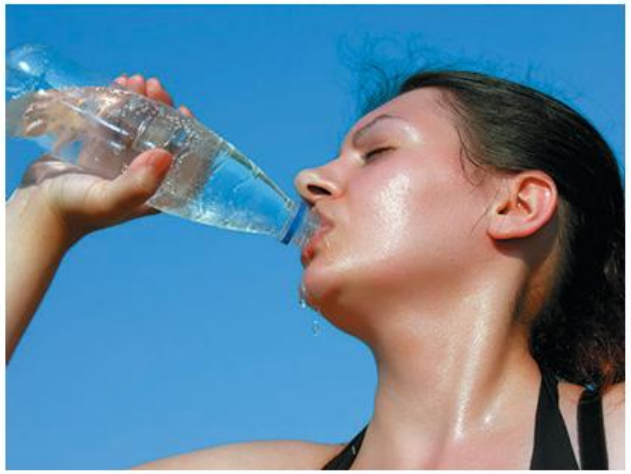
Step 4. Heat of reaction = $-113.0\ kJ \div 0.0868\ mol = -1.30 \times 10^3\ kJ/mol$

5.5 Calorimetry

Chemistry and Life: The Regulation of Body Temperature

Q. How does the body keep its temperature near 37 °C?

- The body is a thermodynamic system: food is oxidized, e.g. $C_6H_{12}O_6(s) + 6O_2(g) \rightarrow 6CO_2(g) + 6H_2O(l)$, $\Delta H = -2803$ kJ.
- About 40% of this energy does work (muscle contraction, nerve activity); the rest is released as heat, partly used to maintain body temperature.
- Heat is lost by **radiation**, **convection** and **evaporation** of sweat: $H_2O(l) \rightarrow H_2O(g)$, $\Delta H = +44.0$ kJ.
- Too hot: more blood flow to the skin and sweating (up to 2–4 L per hour) – fluids must be replaced, or heat exhaustion/heat stroke can follow.
- Too cold: less blood flow to the skin and shivering (muscle work generates heat); failure leads to hypothermia.



▲ Figure 5.20 Perspiration.

Q. Why do we feel more uncomfortable on hot, humid days?

Answer: Evaporation of sweat is slower when the air is already humid, so less heat is removed.

5.5 Calorimetry

Q. How much heat is needed to raise the temperature of 10.0 g of Fe(s) from 25 °C to 75 °C? (C_s = 0.45 J/g·K)

Answer: $q = (0.45\ J/g{\cdot}K)(10.0\ g)(50\ K) = 2.3 \times 10^2\ J$

Q. **(a)** What amount of heat (in joules) is required to raise the temperature of 1 g of water by 1 kelvin? **(b)** What is the heat capacity of 370 g of liquid water? **(c)** How many kJ of heat are needed to raise the temperature of 5.00 kg of liquid water from 24.6 to 46.2 °C?

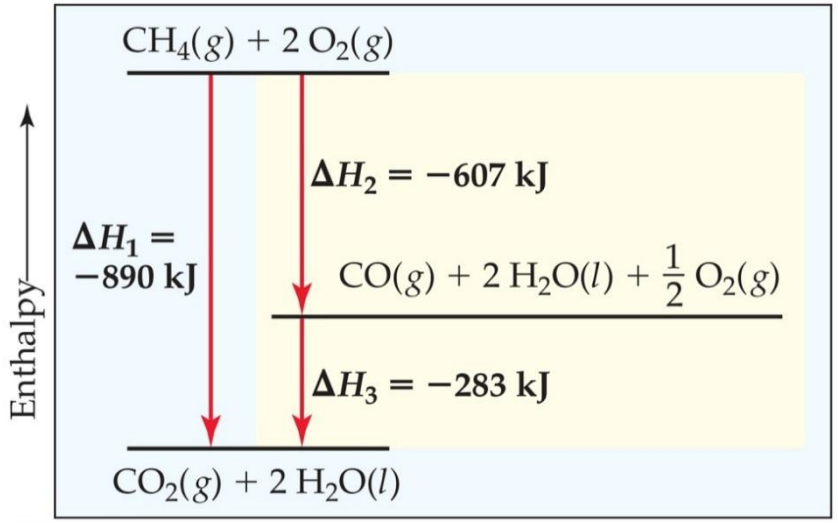
Q. When an 18.6 g sample of solid potassium hydroxide dissolves in 200.0 g of water in a coffee-cup calorimeter (Figure 5.18), the temperature rises from 23.7 to 44.5 °C. **(a)** Calculate the quantity of heat (in kJ) released in the reaction. **(b)** Using your result from part (a), calculate ΔH (in kJ>mol KOH) for the solution process. Assume that the specific heat of the solution is the same as that of pure water.

5.6 Hess's Law

Hess's Law

Q. Can we find ΔH for a reaction that is difficult to measure directly?

- **Hess's law**: if a reaction is carried out in a series of steps, ΔH for the overall reaction equals the **sum** of ΔH for the individual steps.
- It works because H is a **state function**.
- Example (diagram): CH_4 burns to CO_2 directly, or first to CO and then to CO_2.



Enthalpy diagram for combustion of 1 mol of methane.

1. $CH_4(g) + 2O_2(g) \rightarrow CO(g) + 2H_2O(l) + \frac{1}{2}O_2(g)$ $\Delta H_2 = -607$ kJ
2. $CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g)$ $\Delta H_3 = -283$ kJ

Overall: $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$

$\Delta H_1 = \Delta H_2 + \Delta H_3 = -890$ kJ

5.6 Hess's Law

Strategy for Using Hess's Law

Q. How do we combine given equations to obtain a target equation?

1. Write the target equation and note on which side each substance appears.
2. Choose given equations containing those substances. **Reverse** an equation if needed – and change the sign of its ΔH.
3. **Multiply** an equation (and its ΔH) so that the coefficients match the target.
4. **Add** the equations and cancel species that appear on both sides.
5. **Add** the ΔH values. Check that the sum gives exactly the target equation.

5.6 Hess's Law

Example-1

Q. Calculate ΔH for $C(s) + \frac{1}{2}O_2(g) \rightarrow CO(g)$, given:

(1) $C(s) + O_2(g) \rightarrow CO_2(g)$ $\quad \Delta H = -393.5$ kJ

(2) $CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g)$ $\quad \Delta H = -283.0$ kJ

Step 1. Keep (1) – C is a reactant in the target:

$C(s) + O_2(g) \rightarrow CO_2(g)$ $\quad \Delta H = -393.5$ kJ

Step 2. Reverse (2) – CO is a product in the target:

$CO_2(g) \rightarrow CO(g) + \frac{1}{2}O_2(g)$ $\quad \Delta H = +283.0$ kJ

Step 3. Add; CO_2 and $\frac{1}{2}O_2$ cancel:

$C(s) + \frac{1}{2}O_2(g) \rightarrow CO(g)$ $\quad \Delta H = -393.5 + 283.0 =$ **-110.5 kJ**

5.6 Hess's Law

Example-2

Q. Calculate ΔH for $2C(s) + H_2(g) \rightarrow C_2H_2(g)$, given:

(1) $C_2H_2(g) + 5/2\ O_2(g) \rightarrow 2CO_2(g) + H_2O(l)$ $\quad \Delta H = -1299.6$ kJ

(2) $C(s) + O_2(g) \rightarrow CO_2(g)$ $\quad \Delta H = -393.5$ kJ

(3) $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$ $\quad \Delta H = -285.8$ kJ

Step 1. Reverse (1): $2CO_2 + H_2O(l) \rightarrow C_2H_2 + 5/2\ O_2$ $\quad \Delta H = +1299.6$ kJ

Step 2. Multiply (2) by 2: $2C + 2O_2 \rightarrow 2CO_2$ $\quad \Delta H = 2(-393.5) = -787.0$ kJ

Step 3. Keep (3): $H_2 + \frac{1}{2}O_2 \rightarrow H_2O(l)$ $\quad \Delta H = -285.8$ kJ

Step 4. Add ($2CO_2$, H_2O and $5/2\ O_2$ cancel): $2C(s) + H_2(g) \rightarrow C_2H_2(g)$

$\Delta H = 1299.6 - 787.0 - 285.8 =$ **+226.8 kJ**

5.7 Enthalpies of Formation

Equations Associated with Enthalpies of Formation

Q. What is a standard enthalpy of formation?

- **Enthalpy of formation, ΔH_f**: ΔH for forming a compound from its constituent elements.
- **Standard state**: the pure form of a substance at 1 atm and the temperature of interest (usually 298 K = 25 °C).
- **Standard enthalpy of formation, $\Delta H°_f$**: ΔH for forming **1 mol** of a compound from its elements, all in their standard states.
- $\Delta H°_f$ of the **most stable form of an element** is **zero**: e.g. C(graphite), $O_2(g)$, $H_2(g)$, $Br_2(l)$, Na(s).

Example – formation of ethanol:

$2C(\text{graphite}) + 3H_2(g) + \frac{1}{2}O_2(g) \rightarrow C_2H_5OH(l)$ $\quad\quad$ $\Delta H°_f = -277.7$ kJ/mol

5.7 Enthalpies of Formation

Equations Associated with Enthalpies of Formation

TABLE 5.3 Standard Enthalpies of Formation, ΔH°_f, at 298 K

Substance	Formula	ΔH°_f (kJ/mol)	Substance	Formula	ΔH°_f (kJ/mol)
Acetylene	$C_2H_2(g)$	226.7	Hydrogen chloride	$HCl(g)$	−92.30
Ammonia	$NH_3(g)$	−46.19	Hydrogen fluoride	$HF(g)$	−268.60
Benzene	$C_6H_6(l)$	49.0	Hydrogen iodide	$HI(g)$	25.9
Calcium carbonate	$CaCO_3(s)$	−1207.1	Methane	$CH_4(g)$	−74.80
Calcium oxide	$CaO(s)$	−635.5	Methanol	$CH_3OH(l)$	−238.6
Carbon dioxide	$CO_2(g)$	−393.5	Propane	$C_3H_8(g)$	−103.85
Carbon monoxide	$CO(g)$	−110.5	Silver chloride	$AgCl(s)$	−127.0
Diamond	$C(s)$	1.88	Sodium bicarbonate	$NaHCO_3(s)$	−947.7
Ethane	$C_2H_6(g)$	−84.68	Sodium carbonate	$Na_2CO_3(s)$	−1130.9
Ethanol	$C_2H_5OH(l)$	−277.7	Sodium chloride	$NaCl(s)$	−410.9
Ethylene	$C_2H_4(g)$	52.30	Sucrose	$C_{12}H_{22}O_{11}(s)$	−2221
Glucose	$C_6H_{12}O_6(s)$	−1273	Water	$H_2O(l)$	−285.8
Hydrogen bromide	$HBr(g)$	−36.23	Water vapor	$H_2O(g)$	−241.8

Q. Which substances in Table 5.3 have a positive ΔH°_f? What does this tell us?

5.7 Enthalpies of Formation

Equations Associated with Enthalpies of Formation

Q. For which of these reactions at 25 °C does ΔH represent a standard enthalpy of formation?

(a) $2Na(s) + \frac{1}{2}O_2(g) \rightarrow Na_2O(s)$

(b) $2K(l) + Cl_2(g) \rightarrow 2KCl(s)$

(c) $C_6H_{12}O_6(s) \rightarrow 6C(\text{diamond}) + 6H_2(g) + 3O_2(g)$

(a) **Yes**: 1 mol of Na_2O forms from its elements in their standard states.

(b) **No**: K is a solid (not a liquid) at 25 °C, and 2 mol of KCl are formed.

(c) **No**: the compound is a reactant (the reverse of formation), and diamond is not the standard state of carbon.

Q. Write the formation reaction for $CaCO_3(s)$.

Answer: $Ca(s) + C(\text{graphite}) + 3/2\ O_2(g) \rightarrow CaCO_3(s)$ $\quad \Delta H°_f = -1207.1$ kJ/mol

5.7 Enthalpies of Formation

Using Enthalpies of Formation to Calculate Enthalpies of Reaction

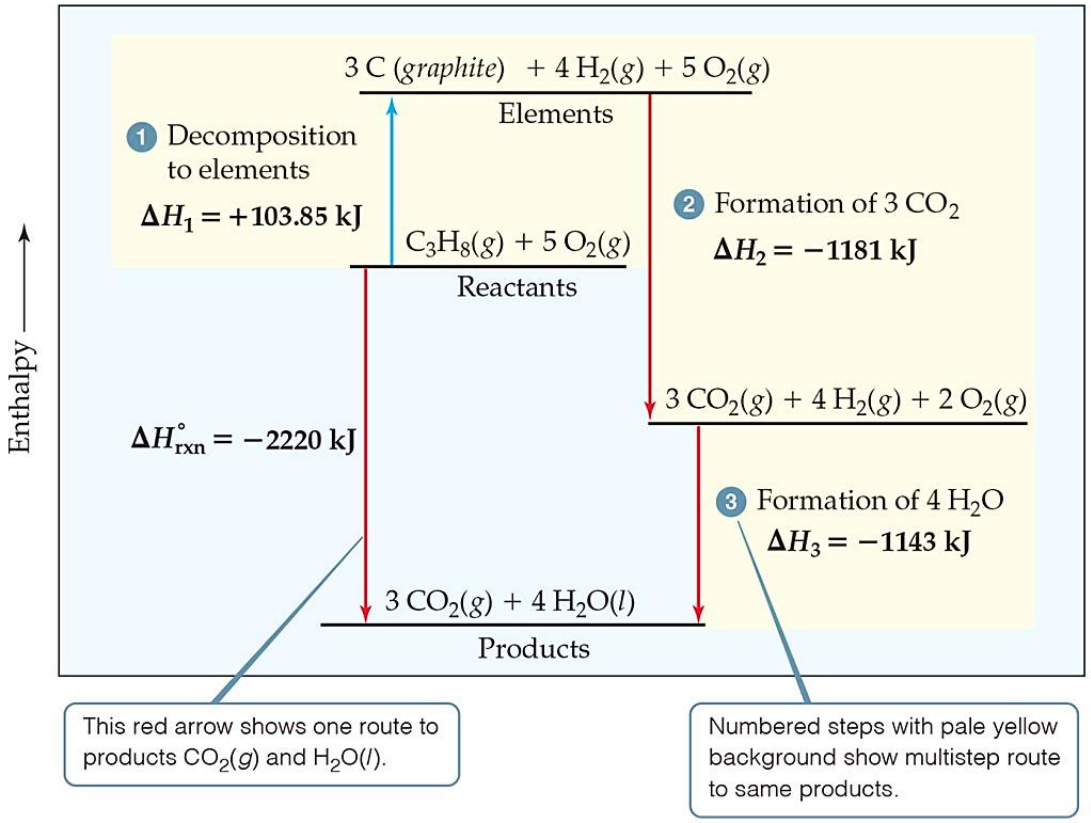
Q. How can $\Delta H°_f$ values give $\Delta H°$ for $C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l)$?

1. Decompose propane into its elements:
   $C_3H_8(g) \rightarrow 3C(s) + 4H_2(g)$
   $\Delta H_1 = -\Delta H°_f(C_3H_8) = +103.85\ kJ$

2. Form 3 mol of CO_2:
   $3C(s) + 3O_2(g) \rightarrow 3CO_2(g)$
   $\Delta H_2 = 3(-393.5\ kJ) = -1180.5\ kJ$

3. Form 4 mol of $H_2O(l)$:
   $4H_2(g) + 2O_2(g) \rightarrow 4H_2O(l)$
   $\Delta H_3 = 4(-285.8\ kJ) = -1143.2\ kJ$

Sum: $\Delta H°_{rxn} = \Delta H_1 + \Delta H_2 + \Delta H_3 = -2219.9\ kJ$



Enthalpy diagram for propane combustion.

5.7 Enthalpies of Formation

Using Enthalpies of Formation to Calculate Enthalpies of Reaction

Q. Is there a shortcut that does all of these steps at once?

$$\Delta H^\circ_{rxn} = \Sigma\, n\Delta H^\circ_f(\text{products}) - \Sigma\, m\Delta H^\circ_f(\text{reactants})$$

- n and m are the stoichiometric coefficients of the balanced equation.
- $\Delta H^\circ_f = 0$ for elements in their standard states (e.g. $O_2(g)$).
- Use the correct physical state: $\Delta H^\circ_f[H_2O(l)] = -285.8$ kJ/mol, but $\Delta H^\circ_f[H_2O(g)] = -241.8$ kJ/mol.
- Common mistakes: forgetting the coefficients, or calculating "reactants – products".

5.7 Enthalpies of Formation

Using Enthalpies of Formation to Calculate Enthalpies of Reaction

Q. Calculate $\Delta H°$ for the combustion of propane, $C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l)$, and of benzene, $C_6H_6(l) + 15/2\ O_2(g) \rightarrow 6CO_2(g) + 3H_2O(l)$. Which releases more heat per gram?

Step 1. Propane: $\Delta H° = [3(-393.5) + 4(-285.8)] - [(-103.85) + 5(0)]$
$= (-2323.7) - (-103.85) =$ **-2219.9 kJ**

Step 2. Benzene: $\Delta H° = [6(-393.5) + 3(-285.8)] - [(49.0) + 15/2(0)]$
$= (-3218.4) - (49.0) =$ **-3267 kJ**

Step 3. Per gram: propane 2220 kJ ÷ 44.1 g = **50.3 kJ/g**; benzene 3267 kJ ÷ 78.1 g = **41.8 kJ/g**

Step 4. Propane releases more heat **per gram**, although benzene releases more **per mole**.

5.7 Enthalpies of Formation

Using Enthalpies of Formation to Calculate Enthalpies of Reaction

Q. The standard enthalpy change for the reaction $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$ is 178.1 kJ. Use Table 5.3 to calculate the standard enthalpy of formation of $CaCO_3(s)$.

Step 1. $\Delta H°_{rxn} = [\Delta H°_f(CaO) + \Delta H°_f(CO_2)] - \Delta H°_f(CaCO_3)$

Step 2. $178.1\ kJ = [(-635.5\ kJ) + (-393.5\ kJ)] - \Delta H°_f(CaCO_3)$

Step 3. $\Delta H°_f(CaCO_3) = -1029.0\ kJ - 178.1\ kJ =$ **-1207.1 kJ/mol**

Check: Table 5.3 lists -1207.1 kJ/mol for $CaCO_3(s)$ – the values agree.

Q. Use Table 5.3 to calculate $\Delta H°$ for $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$.

Answer: **-890.3 kJ**

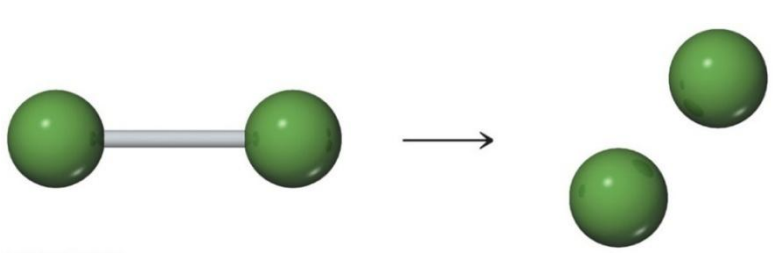
5.8 Bond Enthalpies

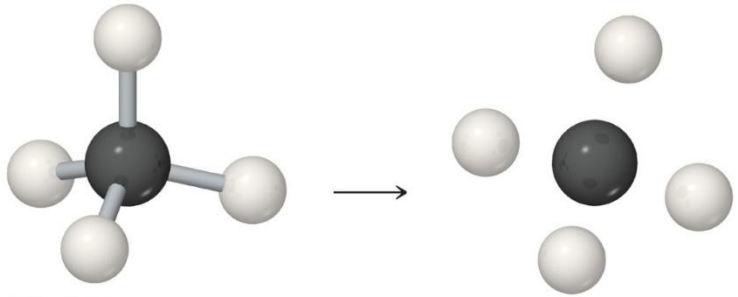
Bond Enthalpies and the Enthalpies of Reactions

Q. To break a chemical bond, must energy be added or removed? What about forming a bond?

> **Answer:** Breaking a bond always **requires** energy (endothermic, $\Delta H > 0$). Forming a bond always **releases** energy (exothermic, $\Delta H < 0$).

Q. How much energy does it take to break a chemical bond?





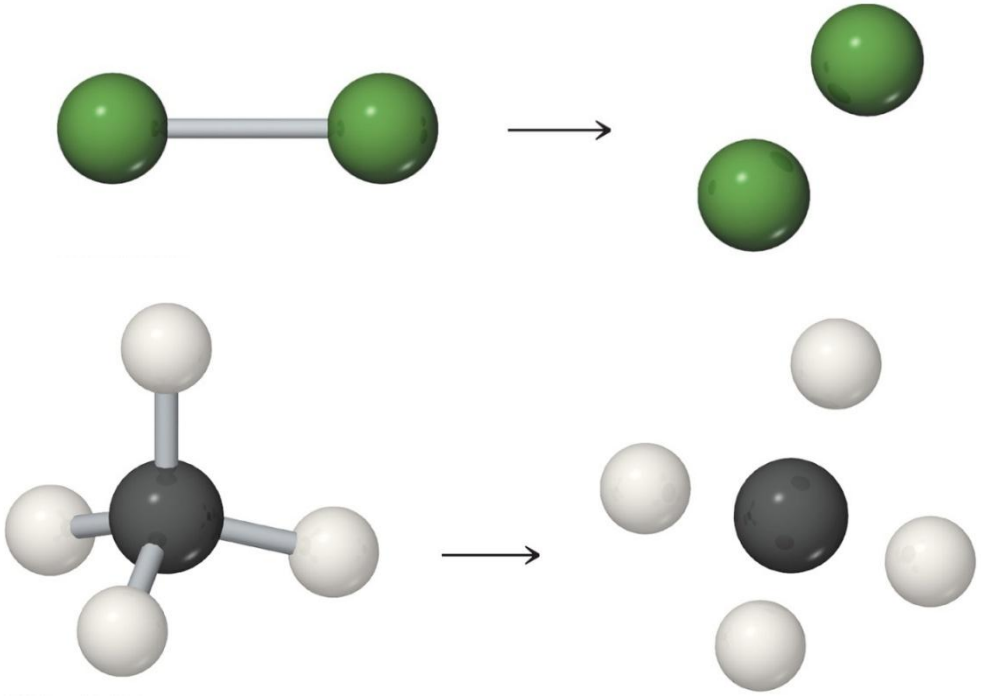
5.8 Bond Enthalpies

Bond Enthalpy

Q. How much energy does it take to break a chemical bond?

- **Bond enthalpy, D**: ΔH for breaking 1 mol of a particular bond in a **gaseous** substance.
- $Cl_2(g) \rightarrow 2Cl(g)$ D(Cl–Cl) = 242 kJ
- $CH_4(g) \rightarrow C(g) + 4H(g)$ ΔH = 1660 kJ

  so the average C–H bond enthalpy is 1660 ÷ 4 = 415 kJ.
- The C–H bond enthalpy varies slightly from molecule to molecule, so tables list **average** values.



5.8 Bond Enthalpies

Bond Enthalpies and the Enthalpies of Reactions

TABLE 5.4 Average Bond Enthalpies (kJ/mol)

C—H	413	N—H	391	O—H	463	F—F	155
C—C	348	N—N	163	O—O	146		
C=C	614	N—O	201	O=O	495	Cl—F	253
C—N	293	N—F	272	O—F	190	Cl—Cl	242
C—O	358	N—Cl	200	O—Cl	203		
C=O	799	N—Br	243	O—I	234	Br—F	237
C—F	485					Br—Cl	218
C—Cl	328	H—H	436			Br—Br	193
C—Br	276	H—F	567				
C—I	240	H—Cl	431			I—Cl	208
		H—Br	366			I—Br	175
		H—I	299			I—I	151

Q. What patterns can you see in Table 5.4?

- Bond enthalpies are always **positive**: energy is needed to break bonds.
- The larger the bond enthalpy, the **stronger** the bond.
- Multiple bonds are stronger than single bonds:

  C–C 348, C=C 614 kJ/mol

  C–O 358, C=O 799 kJ/mol
- Energy is released when bonds form between gaseous atoms.

5.8 Bond Enthalpies

Bond Enthalpies and the Enthalpies of Reactions

Q. How can bond enthalpies tell us whether a reaction is exothermic or endothermic?

- Imagine the reaction in two steps: (1) **break** all the bonds in the reactants (energy in, $\Delta H > 0$); (2) **form** all the bonds in the products (energy out, $\Delta H < 0$).

$$\Delta H_{rxn} \approx \Sigma(\text{bond enthalpies of bonds broken}) - \Sigma(\text{bond enthalpies of bonds formed})$$

- Bonds formed **stronger** than bonds broken $\rightarrow \Delta H < 0$ (exothermic).
- Bonds broken stronger than bonds formed $\rightarrow \Delta H > 0$ (endothermic).
- Applies to **gas-phase** reactions; the result is an **estimate** because tabulated values are averages.

5.8 Bond Enthalpies

Bond Enthalpies and the Enthalpies of Reactions: **Example**

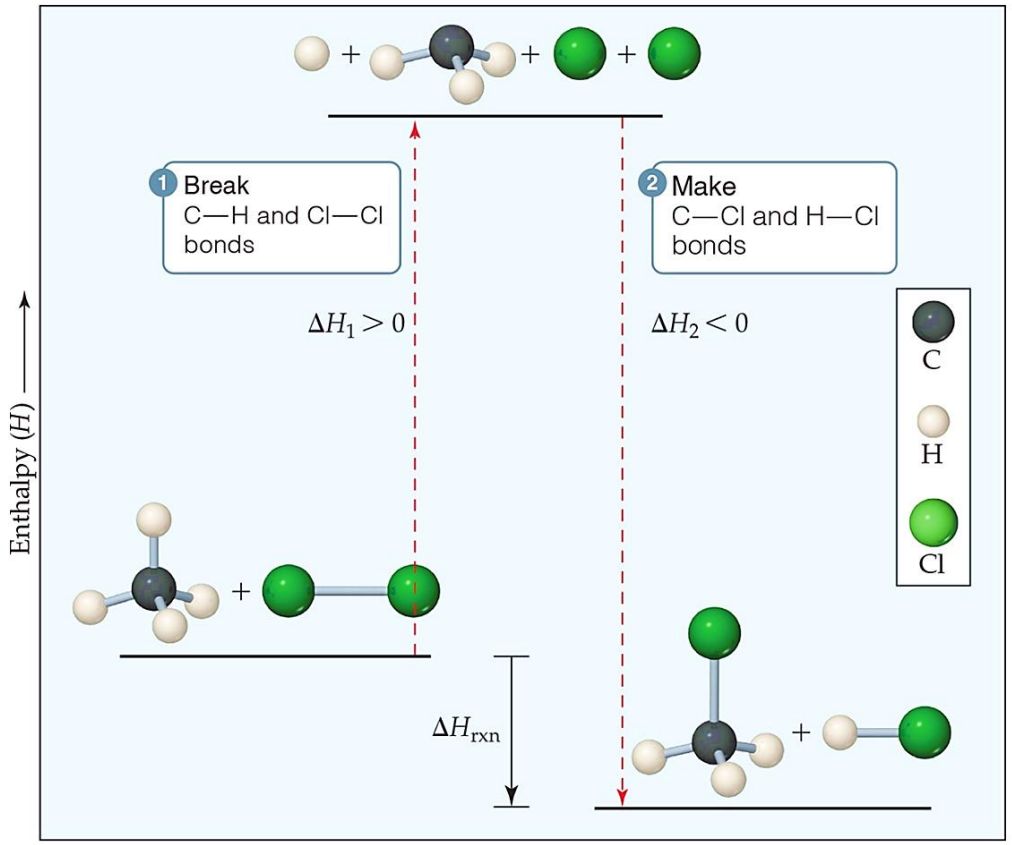
Q. Estimate ΔH for this reaction from bond enthalpies (Table 5.4).

Step 1. Bonds broken: 1 C–H + 1 Cl–Cl
413 + 242 = 655 kJ

Step 2. Bonds formed: 1 C–Cl + 1 H–Cl
328 + 431 = 759 kJ

Step 3. ΔH ≈ 655 kJ – 759 kJ = **–104 kJ**

Step 4. Stronger bonds are formed than are broken → **exothermic**.



Using bond enthalpies to estimate H_{rxn}.

5.8 Bond Enthalpies

Bond Enthalpies and the Enthalpies of Reactions

Q. Use Table 5.4 to estimate ΔH for $2C_2H_6(g) + 7O_2(g) \rightarrow 4CO_2(g) + 6H_2O(g)$. Compare with the value from $\Delta H°_f$ data (–2855 kJ).

Step 1. Bonds broken: 12 C–H + 2 C–C + 7 O=O
12(413) + 2(348) + 7(495) = 9117 kJ

Step 2. Bonds formed: 8 C=O + 12 O–H
8(799) + 12(463) = 11,948 kJ

Step 3. $\Delta H \approx$ 9117 kJ – 11,948 kJ = **–2831 kJ**

Step 4. Close to –2855 kJ from $\Delta H°_f$ data: bond enthalpies give a good **estimate**.

5.8 Bond Enthalpies

Q. Without doing any calculations, predict the sign of ΔH for each of the following reactions:

(a) $NaCl(s) \longrightarrow Na^+(g) + Cl^-(g)$

(b) $2\,H(g) \longrightarrow H_2(g)$

(c) $Na(g) \longrightarrow Na^+(g) + e^-$

(d) $I_2(s) \longrightarrow I_2(l)$

Q. Why are ΔH values calculated from bond enthalpies only estimates?

Answer: Table values are **averages** over many molecules; the actual strength of a bond depends on the rest of the molecule, and the method applies only to gases.

5.9 Foods and Fuels

Foods: Fuel Values of Foods

Q. How much energy does a food provide?

- **Fuel value**: the energy released when 1 g of a material is burned (kJ/g).
- Food fuel values are measured by bomb calorimetry.
- Average fuel values: carbohydrates 17 kJ/g ; fats 38 kJ/g; proteins 17 kJ/g
- Food labels use Calories: 1 Cal = 4.184 kJ.

Q. Why do fats provide more than twice as much energy per gram as carbohydrates?

Answer: Fats contain much less oxygen and more C–H bonds; they are less oxidized, so more energy is released when they burn to CO_2 and H_2O.

5.9 Foods and Fuels

Foods: Fuel Values of Foods

TABLE 5.5 Compositions and Fuel Values of Some Common Foods

	Approximate Composition (% by Mass)			Fuel Value
	Carbohydrate	Fat	Protein	kJ/g
Carbohydrate	100	—	—	17
Fat	—	100	—	38
Protein	—	—	100	17
Apples	13	0.5	0.4	2.5
Beer[a]	1.2	—	0.3	1.8
Bread	52	3	9	12
Cheese	4	37	28	20
Eggs	0.7	10	13	6.0
Fudge	81	11	2	18
Green beans	7.0	—	1.9	1.5
Hamburger	—	30	22	15
Milk (whole)	5.0	4.0	3.3	3.0
Peanuts	22	39	26	23

[a] Beer typically contains 3.5% ethanol, which has fuel value.

Q. Which pure macronutrient listed in the table provides the highest fuel value per gram? Which gives the lowest?

Q. Between bread and fudge, which food has the higher fuel value, and what are their respective carbohydrate percentages by mass

5.9 Foods and Fuels

Foods: Carbohydrates, Fats, and Proteins

Q. Where does the energy in our food come from?

- **Carbohydrates** (17 kJ/g) are broken down into glucose, which is oxidized:

$$C_6H_{12}O_6(s) + 6O_2(g) \rightarrow 6CO_2(g) + 6H_2O(l) \qquad \Delta H° = -2803 \text{ kJ}$$

- **Fats** (38 kJ/g), e.g. tristearin:

$$2C_{57}H_{110}O_6(s) + 163O_2(g) \rightarrow 114CO_2(g) + 110H_2O(l) \qquad \Delta H° = -75{,}520 \text{ kJ}$$

- **Proteins** (17 kJ/g): the reactions in the body are not the same as in a calorimeter (they form different products), so the body obtains less energy than a bomb calorimeter measures.
- Fats are ideal for **energy storage**: highest fuel value and insoluble in water.

5.9 Foods and Fuels

Example

Q. (a) A 28 g serving of breakfast cereal with 120 mL of skim milk provides 8 g protein, 26 g carbohydrate and 2 g fat. Estimate its fuel value. (b) A person uses about 100 Cal per mile when jogging. How many servings provide the energy to jog 3 miles?

Step 1. Protein: 8 g × 17 kJ/g = 136 kJ; carbohydrate: 26 g × 17 kJ/g = 442 kJ; fat: 2 g × 38 kJ/g = 76 kJ

Step 2. Total ≈ **650 kJ**; 650 kJ × (1 Cal / 4.184 kJ) ≈ **160 Cal**

Step 3. 3 miles needs 3 × 100 Cal = 300 Cal → 300 Cal ÷ 160 Cal per serving ≈ **2 servings**

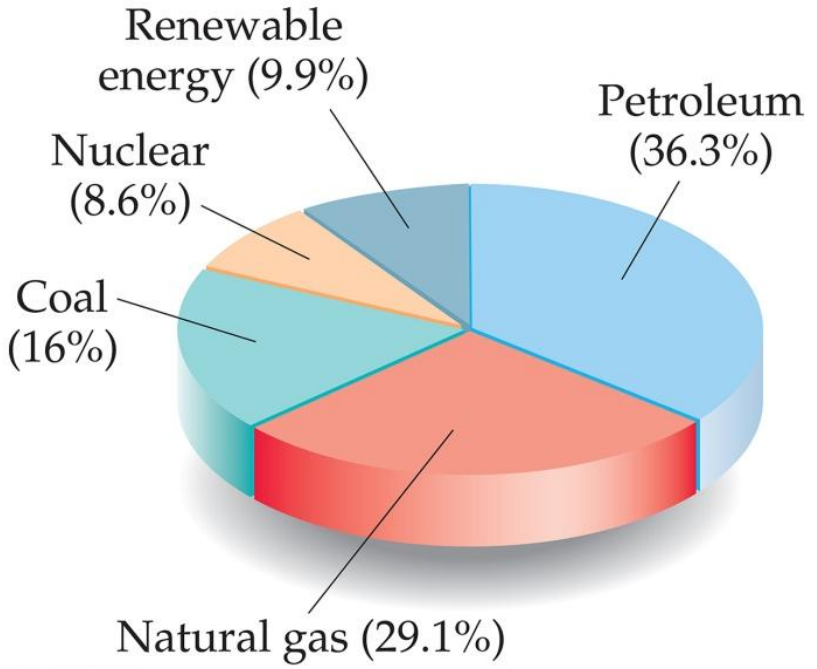
5.9 Foods and Fuels

Fuels: Fuel Values of Common Fuels

Q. Which fuels release the most energy per gram, and where does our energy come from?

TABLE 5.6 Fuel Values and Compositions of Some Common Fuels

	Approximate Elemental Composition (Mass %)			
	C	H	O	Fuel Value (kJ/g)
Wood (pine)	50	6	44	18
Anthracite coal (Pennsylvania)	82	1	2	31
Bituminous coal (Pennsylvania)	77	5	7	32
Charcoal	100	0	0	34
Crude oil (Texas)	85	12	0	45
Gasoline	85	15	0	48
Natural gas	70	23	0	49
Hydrogen	0	100	0	142



- Fuel value increases with **hydrogen content** (and decreases with oxygen content); H_2 has the highest value, 142 kJ/g.
- **Fossil fuels** (petroleum, natural gas and coal) supply about 81% of U.S. energy.

5.9 Foods and Fuels

Other Energy Sources

Q. What are the alternatives to fossil fuels?

- ✓ **Nuclear energy**: 8.6% of U.S. energy needs.
- ✓ **Renewable energy** – solar, wind, geothermal, hydroelectric and biomass: 9.9%.
- ✓ **Biomass** (e.g. sugarcane → ethanol) is renewable because plants regrow, absorbing CO_2.
- ✓ Fossil fuels are **nonrenewable**, and burning them releases CO_2.



Q. Why is hydrogen attractive as a future fuel, and what is its main drawback?

Answer: Highest fuel value (142 kJ/g) and its only combustion product is water; but H_2 must first be produced using energy, and it is difficult to store.

5.9 Foods and Fuels

Chemistry Put to Work: Biofuels

Q. Can fuels made from plants replace petroleum?

- More than 80% of world energy comes from nonrenewable fossil fuels; **biofuels** are liquid fuels derived from biological matter.
- **Bioethanol** is made by fermenting plant carbohydrates: $C_6H_{12}O_6(s) \rightarrow 2C_2H_5OH(l) + 2CO_2(g)$, $\Delta H = +15.8$ kJ. Its fuel value is about two-thirds that of gasoline.
- Corn ethanol (USA) is energy-intensive: only 1.34 J produced per 1.00 J used (34% return) and it competes with food ("food versus fuel").
- Sugarcane ethanol (Brazil): about 8.0 J per 1.0 J used. Cellulosic ethanol from non-food plants and **biodiesel** (soybean, canola, waste oils) are other options.
- Plants store solar energy by photosynthesis: $6CO_2(g) + 6H_2O(l) + \text{sunlight} \rightarrow C_6H_{12}O_6(s) + 6O_2(g)$.

Q. A granola bar contains 20 g carbohydrate, 4 g protein and 6 g fat. Estimate its energy in kJ and in Cal.

Answer: 20(17) + 4(17) + 6(38) = 636 kJ ≈ **640 kJ ≈ 150 Cal**

Chapter 5 Summary

By now, you should be able to:

- ✓ Describe kinetic and potential energy and convert between energy units.
- ✓ Apply the first law of thermodynamics: $\Delta E = q + w$, with the correct signs.
- ✓ Calculate P–V work and explain why $\Delta H = q$ at constant pressure.
- ✓ Use thermochemical equations and stoichiometry to calculate heat flow.
- ✓ Analyse coffee-cup and bomb calorimetry data.
- ✓ Combine thermochemical equations using Hess's law.
- ✓ Calculate ΔH°_{rxn} from standard enthalpies of formation.
- ✓ Estimate ΔH from average bond enthalpies.
- ✓ Compare the fuel values of foods and fuels.

Chapter 5 Summary

Key relationships

Topic	Key relationship
Energy	$E_k = \frac{1}{2}mv^2$; $E_{el} = \kappa Q_1 Q_2/d$; 1 cal = 4.184 J; 1 Cal = 4.184 kJ
First law	$\Delta E = q + w$; energy into the system is positive
P–V work	$w = -P\Delta V$ (1 kPa·L = 1 J = 1 Pa·m^3)
Enthalpy	$H = E + PV$; $\Delta H = q_P$; $\Delta H < 0$ exothermic, $\Delta H > 0$ endothermic
Calorimetry	$q = C_s \times m \times \Delta T$; $q_{rxn} = -q_{soln}$; $q_{rxn} = -C_{cal} \times \Delta T$
Hess's law	$\Delta H_{overall} = \Sigma\ \Delta H_{steps}$
Formation	$\Delta H°_{rxn} = \Sigma\ n\Delta H°_f(\text{products}) - \Sigma\ m\Delta H°_f(\text{reactants})$
Bond enthalpies	$\Delta H_{rxn} \approx \Sigma\ D(\text{bonds broken}) - \Sigma\ D(\text{bonds formed})$
Fuel values	carbohydrate 17 kJ/g, protein 17 kJ/g, fat 38 kJ/g

End-of-Chapter Exercises

From the Textbook

Q. (a) What is the electrostatic potential energy (in joules) between two electrons that are separated by 460 pm? (b) What is the change in potential energy if the distance separating the two electrons is increased to 1.0 nm? (c) Does the potential energy of the two particles increase or decrease when the distance is increased to 1.0 nm? *[Exercise 5.61]*

Q. For the following processes, calculate the change in internal energy of the system and determine whether the process is endothermic or exothermic: (a) A balloon is cooled by removing 0.655 kJ of heat. It shrinks on cooling, and the atmosphere does 382 J of work on the balloon. (b) A 100.0 g bar of gold is heated from 25 °C to 50 °C during which it absorbs 322 J of heat. Assume the volume of the gold bar remains constant. *[Exercise 5.68]*

Q. How much work (in J) is involved in a chemical reaction if the volume decreases from 33.6 L to 11.2 L against a constant pressure of 90.5 kPa? *[Exercise 5.71]*

End-of-Chapter Exercises

From the Textbook

Q. A gas is confined to a cylinder under constant atmospheric pressure. When 0.49 kJ of heat is added to the gas, it expands and does 214 J of work on the surroundings. What are the values of ΔH and ΔE for this process? *[Exercise 5.75]*

Q. Consider the reaction $2CH_3OH(g) \rightarrow 2CH_4(g) + O_2(g)$, $\Delta H = +252.8$ kJ. (a) Is this reaction exothermic or endothermic? (b) Calculate the amount of heat transferred when 24.0 g of $CH_3OH(g)$ is decomposed at constant pressure. (c) For a given sample of CH_3OH, the enthalpy change is 82.1 kJ. How many grams of methane gas are produced? (d) How many kilojoules of heat are released when 38.5 g of $CH_4(g)$ reacts completely with $O_2(g)$ to form $CH_3OH(g)$ at constant pressure? *[Exercise 5.78]*

Q. Consider the decomposition of liquid benzene to gaseous acetylene: $C_6H_6(l) \rightarrow 3C_2H_2(g)$, $\Delta H = +630$ kJ. (a) What is the enthalpy change for the reverse reaction? (b) What is ΔH for the formation of 1 mol of acetylene? (c) Which is more likely to be thermodynamically favored, the forward or the reverse reaction? (d) If $C_6H_6(g)$ were consumed instead of $C_6H_6(l)$, would the magnitude of ΔH increase, decrease or stay the same? *[Exercise 5.80]*

End-of-Chapter Exercises

From the Textbook

Q. (a) When an 8.50 *g* sample of solid ammonium nitrate (NH_4NO_3) dissolves in 120.0 *g* of water in a coffee-cup calorimeter, the temperature drops from 24.0 to 18.9 °*C*. Calculate ΔH (in kJ/mol NH_4NO_3) for the solution process. Assume that the specific heat of the solution is the same as that of pure water. (b) Is this process endothermic or exothermic? *[Exercise 5.84]*

Q. *A* 2.20 *g* sample of phenol (C_6H_5OH) was burned in a bomb calorimeter whose total heat capacity is 11.90 kJ/°*C*. The temperature of the calorimeter plus contents increased from 21.50 to 27.50 °*C*. (a) Write a balanced chemical equation for the bomb calorimeter reaction. (b) What is the heat of combustion per gram of phenol and per mole of phenol? *[Exercise 5.85]*

Q. Under constant-volume conditions, the heat of combustion of naphthalene ($C_{10}H_8$) is 40.18 kJ/*g*. *A* 2.50 *g* sample is burned in a bomb calorimeter; the temperature increases from 21.50 to 28.83 °*C*. (a) What is the total heat capacity of the calorimeter? (b) *A* 1.50 *g* sample of a new organic substance is combusted in the same calorimeter; the temperature increases from 21.14 to 25.08 °*C*. What is its heat of combustion per gram? (c) If some water in the calorimeter were lost, how would this change the heat capacity of the calorimeter? *[Exercise 5.86]*

End-of-Chapter Exercises

From the Textbook

Q. From the enthalpies of reaction $2C(s) + O_2(g) \rightarrow 2CO(g)$, $\Delta H = -221.0$ kJ and $2C(s) + O_2(g) + 4H_2(g) \rightarrow 2CH_3OH(g)$, $\Delta H = -402.4$ kJ, calculate ΔH for the reaction $CO(g) + 2H_2(g) \rightarrow CH_3OH(g)$. *[Exercise 5.88]*

Q. Given $N_2(g) + O_2(g) \rightarrow 2NO(g)$, $\Delta H = +180.7$ kJ; $2NO(g) + O_2(g) \rightarrow 2NO_2(g)$, $\Delta H = -113.1$ kJ; $2N_2O(g) \rightarrow 2N_2(g) + O_2(g)$, $\Delta H = -163.2$ kJ, use Hess's law to calculate ΔH for the reaction $N_2O(g) + NO_2(g) \rightarrow 3NO(g)$. *[Exercise 5.89]*

Q. Using values from Appendix C, calculate ΔH for each reaction: (a) $CaO(s) + 2HF(g) \rightarrow CaF_2(s) + H_2O(g)$; (b) $Fe_2O_3(s) + 3C(s) \rightarrow 2Fe(s) + 3CO(g)$; (c) $2CO(g) + 2NO(g) \rightarrow N_2(g) + 2CO_2(g)$; (d) $4NH_3(g) + 5O_2(g) \rightarrow 4NO(g) + 6H_2O(g)$. *[Exercise 5.93]*

End-of-Chapter Exercises

From the Textbook

Q. The complete combustion of 1 mol of diethyl ether, $C_4H_{10}O(l)$, to $CO_2(g)$ and $H_2O(l)$ yields $\Delta H° = -2723.7$ kJ. (a) Write a balanced equation for the combustion of 1 mol of $C_4H_{10}O(l)$. (b) Using this information and data in Table 5.3, calculate $\Delta H°_f$ for diethyl ether. *[Exercise 5.95]*

Q. Consider the reaction $H_2(g) + Br_2(l) \rightarrow 2HBr(g)$. (a) Use the bond enthalpies in Table 5.4 to estimate ΔH, ignoring the fact that bromine is in the liquid state. (b) Without doing a calculation, predict whether your estimate is more negative or less negative than the true reaction enthalpy. (c) Use the enthalpies of formation in Appendix C to determine the true reaction enthalpy. *[Exercise 5.100]*

Q. A hamburger contains 28 g fat, 46 g carbohydrate and 25 g protein. What is the fuel value in kJ in one hamburger? How many Calories does it provide? *[Exercise 5.102]*